

EECE 305: FUNDAMENTALS OF ELECTROMAGNETICS

LECTURE NOTES

SU YAN, PH.D.

Associate Professor
Department of Electrical Engineering and Computer Science
Howard University, Washington, DC 20059-1015, USA

su.yan@howard.edu



Outline I

- 1 Course Information & Requirements
- 2 Chapter 1: Introduction
- 3 Chapter 2: Electrostatics
 - Section 2.1: Vectors in the Cartesian Coordinate System
 - Section 2.2: Coulomb's Law
 - Section 2.3: The Spherical Coordinate System
 - Section 2.4: Line Charge and the Cylindrical Coordinate System
 - Section 2.5: Surface and Volume Charge
 - Section 2.6: Electric Flux Density
 - Section 2.7: Gauss's Law
 - Section 2.8: Divergence and the Point Form of Gauss's Law
 - Section 2.9: Electric Potential
 - Section 2.10: Conductors and Ohm's Law
 - Section 2.11: Dielectrics
 - Section 2.12: Boundary Conditions
 - Section 2.13: Capacitance
- 4 Chapter 3: Magnetostatics
 - Section 3.1: Magnetic Fields and The Cross Product
 - Section 3.2: Biot-Savart's Law
 - Section 3.3: Ampère's Circuital Law
 - Section 3.4: Curl and the Point Form of Ampère's Circuital Law
 - Section 3.5: Magnetic Flux Density
 - Section 3.6: Magnetic Forces
 - Section 3.7: Magnetic Materials
 - Section 3.8: Boundary Conditions
 - Section 3.9: Inductance and Magnetic Energy



Outline II

5 Chapter 4: Dynamic Fields

- Section 4.1: Current Continuity and Relaxation Time
- Section 4.2: Wave Fundamentals
- Section 4.3: Faraday's Law and Transformer EMF
- Section 4.4: Faraday's Law and Motional EMF
- Section 4.5: Displacement Current
- Section 4.6: Maxwell's Equations
- Section 4.7: Lossless TEM Waves
- Section 4.8: Time-Harmonic Fields and Phasors

6 Chapter 5: Plane Waves

- Section 5.1: General Wave Equations
- Section 5.2: Propagation in Lossless, Charge-Free Media
- Section 5.3: Propagation in Dielectrics
- Section 5.4: Propagation in Conductors
- Section 5.5: The Poynting Theorem and Power Transmission
- Section 5.6: Polarization
- Section 5.7: Reflection and Transmission at Normal Incidence
- Section 5.8: Reflection and Transmission at Oblique Incidence
- Chapter 5 Summary

7 Course Review



EECE 305: Fundamentals of Electromagnetics

Course Information

- Credit Hours: 4
- Time: Monday & Wednesday 9:40am – 11:00am
Monday 12:10pm – 1:00pm
- Room: L. K. Downing Building, Rm 3116
- Office Hours: Monday 3:45pm – 5:00pm, or by email appointment.

Instructor Information

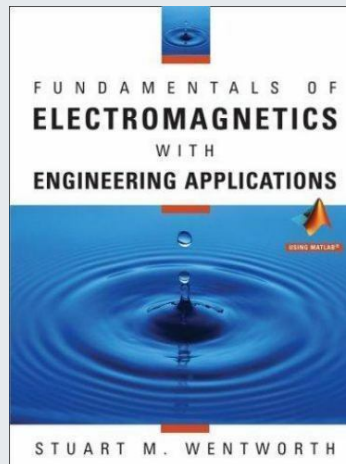
- Dr. Su Yan
- Office: L. K. Downing Building, Ste 3030C
- Phone: (202) 806-2215
- Email: su.yan@howard.edu
- Website: www.sudoem.org



EECE 305: Fundamentals of Electromagnetics

Textbook

- Stuart M. Wentworth, *Fundamentals of Electromagnetics with Engineering Applications*, John Wiley & Sons, 2005.

HOWARD
UNIVERSITY

EECE 305: Fundamentals of Electromagnetics

Course Grading

- | | |
|----------------------------|-----|
| ■ Homework: | 20% |
| ■ Midterm Exams (2 exams): | 30% |
| ■ Project: | 20% |
| ■ Final Exam: | 30% |

Grading Scale

- | | |
|-------------|---|
| ■ 90 – 100: | A |
| ■ 80 – 89: | B |
| ■ 70 – 79: | C |
| ■ 60 – 69: | D |
| ■ Below 60: | F |

HOWARD
UNIVERSITY

EECE 305: Fundamentals of Electromagnetics

Homework

- Once a week: Post on Canvas every Wednesday.
- Due in a week: The following Wednesday in class.
- Late submission: A deduction of 10%.
- Plagiarism: Zero credit.

Midterms and Final Exam

- Close book.
- Midterm 1: On or about Monday, September 21, 9:40am – 11:40am.
- Midterm 2: On or about Monday, October 26, 9:40am – 11:40am.
- Final Exam: Wednesday, December 9, 2:00pm – 4:00pm.

Project

- An essay discussing one of the contemporary topics related to electromagnetics.
- Work in teams.



Course Overview

Electromagnetics (Electromagnetism)

- A branch of physics involving the study of the electromagnetic (EM, EMag) force, a type of physical interaction that occurs between electrically charged particles.
Fundamental physical interactions/forces:
 - 1 Electromagnetic interaction: Long-range
Through electric and magnetic fields and lights (optical fields): EM force $\rightarrow$ Lorentz force; Optical force $\rightarrow$ Optical tweezer
 - 2 Gravitational interaction: Long-range
 - 3 Weak interaction: interaction between subatomic particles
 - 4 Strong interaction: the mechanism responsible for the strong nuclear force
- An important foundation for engineering applications:
 - 1 Basis of electrical/electronic circuits: Circuit theory
 - 2 Wireless communication: Antenna, Radar, Wifi, NFC, etc
 - 3 Power transmission: wired, wireless
 - 4 Imaging and remote sensing: Non-destructive detection, ground penetrating radar, through-wall imaging
 - 5 and many more.



Course Contents

Electrostatics

- Coulomb's law
- Gauss's law
- Ohm's law

Magnetostatics

- Biot-Savart's law
- Ampère's law

Electromagnetics: Dynamic Fields

- Faraday's law
- Maxwell–Ampère's law
- Maxwell's equations
- Plane waves: Propagation, transmission, and reflection



Introduction

Maxwell's Equations:

- Faraday's law:

$$\nabla \times \vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

- Maxwell–Ampère's law:

$$\nabla \times \vec{H} = \frac{\partial \vec{D}}{\partial t} + \vec{J}$$

- Gauss's law:

$$\nabla \cdot \vec{D} = \rho$$

- Gauss's law – Magnetic case:

$$\nabla \cdot \vec{B} = 0$$

Charge conservation law:

$$\nabla \cdot \vec{J} = -\frac{\partial \rho}{\partial t}$$



Introduction

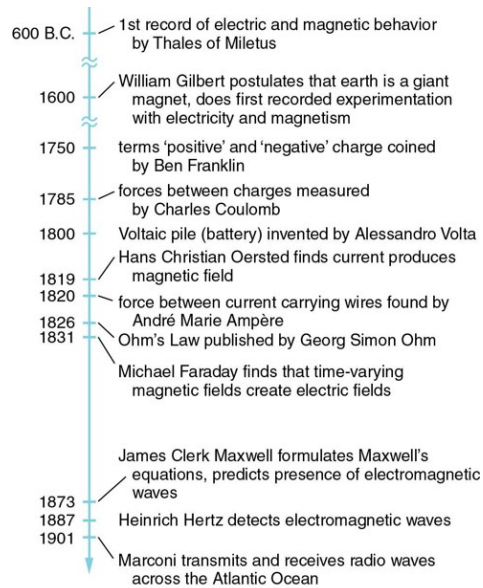


Figure: The key historical events in electromagnetics.

Introduction

Electromagnetic field are generated

Naturally:

- Solar radiation
- Lightning

Artificially:

- Radio stations
- Cell phones
- Power lines

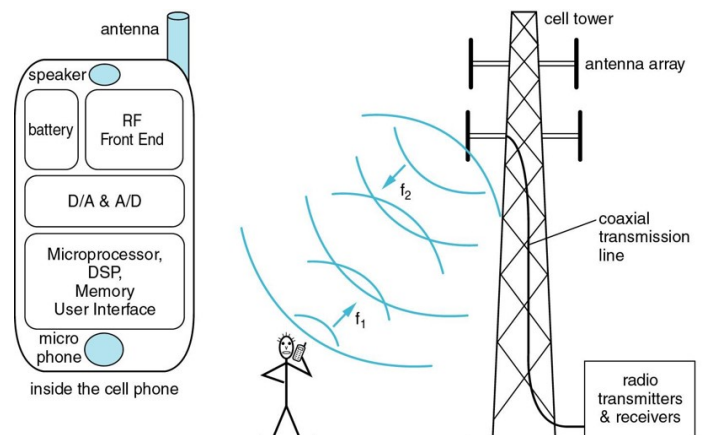


Figure: Typical cellular telephone system.

Electromagnetic Spectrum

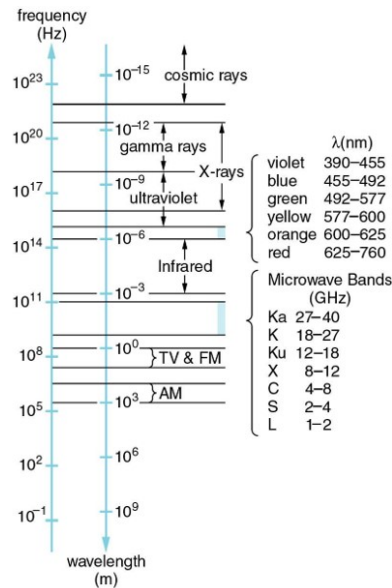


Figure: The electromagnetic spectrum.

Engineering Notation Prefixes

Table 1.1 Engineering Notation Prefixes

Prefix	Pronunciation	Multiplier
T	tera	10 ¹²
G	giga	10 ⁹
M	mega	10 ⁶
k	kilo	10 ³
m	milli	10 ⁻³
μ	micro	10 ⁻⁶
n	nano	10 ⁻⁹
p	pico	10 ⁻¹²
f	femto	10 ⁻¹⁵

Figure: Engineering Notation Prefixes.

Cartesian Coordinate System

Cartesian Coordinate System

A right-handed system of orthogonal coordinate axes: x , y , and z .

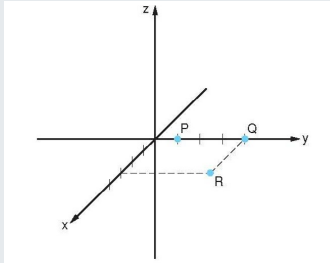


Figure: Cartesian coordinate points $P(0, 1, 0)$, $Q(0, 4, 0)$, and $R(3, 4, 0)$.

- **Orthogonal:** Each axis is perpendicular to the other two.
- **Right-handed:** Curl fingers of the right hand from x to y axes, the thumb points to the z direction.
- Vectors in Cartesian coordinate system:
 $\vec{P} = (0, 1, 0)$, $\vec{Q} = (0, 4, 0)$, $\vec{R} = (3, 4, 0)$, and $\vec{O} = (0, 0, 0)$ (origin).
- Vector $\vec{P}$: magnitude: 1, direction: $\hat{y}$ direction $\rightarrow$ a unit vector



HOWARD
UNIVERSITY

Cartesian Coordinate System

Unit Vector

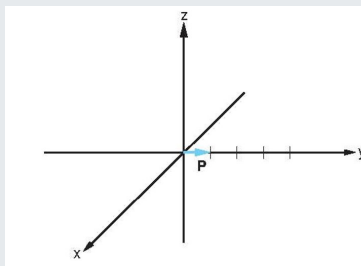


Figure: A unit vector $\vec{P} = \hat{y}$.

- Define unit vectors in x , y , and z directions:
 $\hat{x} = (1, 0, 0)$, $\hat{y} = (0, 1, 0)$, and $\hat{z} = (0, 0, 1)$
- With the definition of unit vectors:
 $\vec{P} = (0, 1, 0) = 1\hat{y}$, $\vec{Q} = (0, 4, 0) = 4\hat{y}$, $\vec{R} = (3, 4, 0) = 3\hat{x} + 4\hat{y}$.
- Alternatively: $\vec{R} = (3, 4, 0) = 3\hat{x} + 4\hat{y} = 5\vec{R} = R\hat{R}$ using Pythagorean theorem:
 Magnitude: $R = |\vec{R}| = \sqrt{3^2 + 4^2} = 5$;
 Direction: denoted by the unit vector: $\hat{R} = \frac{\vec{R}}{R} = \frac{3}{5}\hat{x} + \frac{4}{5}\hat{y}$.



HOWARD
UNIVERSITY

Vector Calculation

■ Summation & subtraction:

$$\vec{A} = A_x \hat{x} + A_y \hat{y} + A_z \hat{z}$$

$$\vec{B} = B_x \hat{x} + B_y \hat{y} + B_z \hat{z}$$

$$\vec{A} \pm \vec{B} = (A_x \pm B_x) \hat{x} + (A_y \pm B_y) \hat{y} + (A_z \pm B_z) \hat{z}$$

■ Scalar-vector product & division:

$$c\vec{A} = cA_x \hat{x} + cA_y \hat{y} + cA_z \hat{z}$$

Example 2.1

Given $\vec{P} = (0, -4, 0)$, $\vec{Q} = (0, 0, 5)$, $\vec{R} = (1, 0, 0)$, and $\vec{S} = (0, 5, 0)$, find

- (a) the vector $\vec{A}$ from $\vec{P}$ to $\vec{Q}$;
- (b) the vector $\vec{B}$ from $\vec{R}$ to $\vec{S}$;
- (c) the direction of $\vec{A} + \vec{B}$.



Vector Calculation

Example 2.1

Given $\vec{P} = (0, -4, 0)$, $\vec{Q} = (0, 0, 5)$, $\vec{R} = (1, 0, 0)$, and $\vec{S} = (0, 5, 0)$, find

- (a) the vector $\vec{A}$ from $\vec{P}$ to $\vec{Q}$:

$$\vec{A} = \vec{Q} - \vec{P} = (0, 4, 5) = 4\hat{y} + 5\hat{z}$$

- (b) the vector $\vec{B}$ from $\vec{R}$ to $\vec{S}$:

$$\vec{B} = \vec{S} - \vec{R} = (-1, 5, 0) = -\hat{x} + 5\hat{y}$$

- (c) the direction of $\vec{A} + \vec{B}$:

$$\vec{A} + \vec{B} = (4\hat{y} + 5\hat{z}) + (-\hat{x} + 5\hat{y}) = -\hat{x} + 9\hat{y} + 5\hat{z}$$

Direction of $\vec{A} + \vec{B}$ is the unit vector:

$$\hat{r} = \frac{\vec{A} + \vec{B}}{|\vec{A} + \vec{B}|} = \frac{-\hat{x} + 9\hat{y} + 5\hat{z}}{\sqrt{(-1)^2 + 9^2 + 5^2}} = -0.097\hat{x} + 0.87\hat{y} + 0.48\hat{z}$$



Differential Elements

Cartesian coordinate system

A differential element is defined by introducing a small increment at a point.

In Cartesian coordinate system: $\Delta x \rightarrow dx$, $\Delta y \rightarrow dy$, $\Delta z \rightarrow dz$.

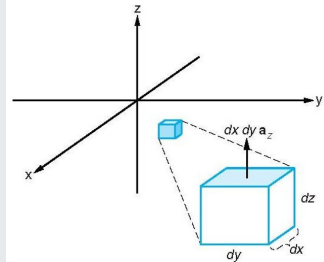


Figure: A differential volume in Cartesian coordinates shows an expanded inset with sides dx , dy , and dz . The differential surface vector $d\vec{S} = \hat{z}dx dy$ is shaded.

- Differential lines: $d\vec{l} = dx\hat{x}, dy\hat{y}, \text{ or } dz\hat{z}$ $d\vec{l} = \hat{x}dx, \hat{y}dy, \text{ or } \hat{z}dz$
- Differential surfaces: $dS = dxdy, dx dz, \text{ or } dy dz$ $d\vec{S} = \pm \hat{z}dx dy, \pm \hat{y}dx dz, \text{ or } \pm \hat{x}dy dz$
- Differential volumes: $dV = dxdydz$



Coulomb's Law

Electrostatic force

- The Greek philosopher Thales:
Amber, after being rubbed on silk, can pick up pieces of lint or straw.
Due to static electricity:
Positive charges induced on amber attract negative charges on lint.
- In 1785, Charles Coulomb measured the force between two charged particles experimentally.

The force between two charged particles is

- 1 Proportional to the product of the two charges;
- 2 Inversely proportional to the square of the distance between the charges;
- 3 Acts in the line containing the two charges;
- 4 Repulsive if both charges are of like sign;
Attractive if the charges are of different signs;
- 5 The proportionality constant: $\frac{1}{4\pi\epsilon_0}$, where $\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m} \approx \frac{10^{-9}}{36\pi} \text{ F/m}$ is the free space permittivity.

— **Coulomb's Law**



Coulomb's Law

Coulomb's Law

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

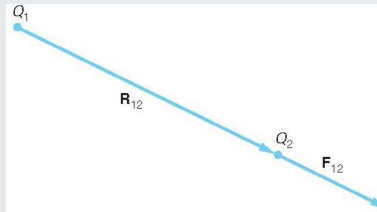


Figure: Coulomb's law example showing the vector force $\vec{F}_{12}$ acting on Q_2 from Q_1 .

- 1 $\vec{F}_{12}$: The force exerted by charge Q_1 on charge Q_2 in Newtons (N).
- 2 Q_1, Q_2 : Charge with signed value in Coulombs (C): $Q > 0$: positive charge; $Q < 0$: negative charge.
- 3 R_{12} : Distance between Q_1 and Q_2 : $R_{12} = |\vec{R}_{12}|$ in meters (m).
- 4 $\hat{a}_{12}$: Unit vector from Q_1 to Q_2 , since it's the force from Q_1 to Q_2 .



Coulomb's Law

Coulomb's Law — Let's check

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

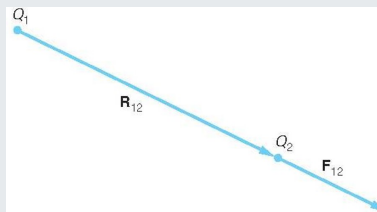


Figure: Coulomb's law example showing the vector force $\vec{F}_{12}$ acting on Q_2 from Q_1 .

- 1 $Q_1 > 0$ and $Q_2 > 0$, or $Q_1 < 0$ and $Q_2 < 0$: $Q_1 Q_2 > 0 \rightarrow \vec{F}_{12} = (+)\hat{a}_{12}$: Repulsive force.
- 2 $Q_1 > 0$ and $Q_2 < 0$, or $Q_1 < 0$ and $Q_2 > 0$: $Q_1 Q_2 < 0 \rightarrow \vec{F}_{12} = (-)\hat{a}_{12}$: Attractive force.



Coulomb's Law

Coulomb's Law — Units

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

Dimensional analysis:

$$[\vec{F}_{12}] = \frac{\text{C} \cdot \text{C}}{\frac{\text{F}}{\text{m}} \cdot \text{m}^2} = \frac{\text{C}^2}{\text{F} \cdot \text{m}}$$

Note

$$F = \frac{C}{V}, \quad C = \frac{J}{V}, \quad \text{and} \quad J = \text{N} \cdot \text{m}$$

it can be shown that

$$[\vec{F}_{12}] = \text{N}$$



Coulomb's Law

Example 2.2

Find the force exerted by $Q_1(0, 0, 4 \text{ m}) = 10 \text{ nC}$ on $Q_2(0, 4 \text{ m}, 0) = 2 \text{ nC}$.

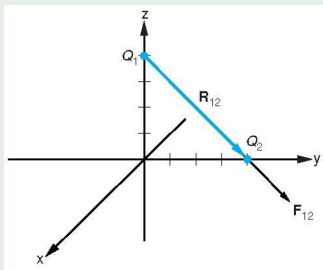


Figure: Coulomb's law example where $Q_1(0, 0, 4 \text{ m}) = 10 \text{ nC}$ and $Q_2(0, 4 \text{ m}, 0) = 2 \text{ nC}$.

■ Solution:

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12}$$

$$Q_1 = 10 \text{ nC} = 10 \times 10^{-9} \text{ C} = 10^{-8} \text{ C}$$

$$Q_2 = 2 \text{ nC} = 2 \times 10^{-9} \text{ C}$$

$$\vec{R}_{12} = (0, 4, 0) - (0, 0, 4) = 4\hat{y} - 4\hat{z}$$

$$R_{12} = |\vec{R}_{12}| = 4\sqrt{2} \text{ m} \Rightarrow R_{12}^2 = 32 \text{ m}^2$$

$$\hat{a}_{12} = \frac{\vec{R}_{12}}{|\vec{R}_{12}|} = \frac{1}{\sqrt{2}}\hat{y} - \frac{1}{\sqrt{2}}\hat{z}$$

$$\epsilon_0 = 8.854 \times 10^{-12} \text{ F/m} \approx \frac{10^{-9}}{36\pi} \text{ F/m} \Rightarrow 4\pi\epsilon_0 \approx \frac{10^{-9}}{9} \text{ F/m}$$

$$\vec{F}_{12} \approx 4.0(\hat{y} - \hat{z}) \text{ nN}$$



Coulomb's Law

Force due to multiple charges

Coulomb's law + Principle of superposition

$$\text{Total force acting on charge } i = \sum_j \text{Force from charge } j \text{ to charge } i$$

Example 2.3

Find the force exerted by $Q_1(0, 0, 4 \text{ m}) = 10 \text{ nC}$ and $Q_3(0, 0, -4 \text{ m}) = 10 \text{ nC}$ on $Q_2(0, 4 \text{ m}, 0) = 2 \text{ nC}$.

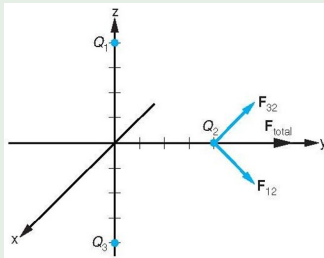


Figure: Vector addition is used to combine the coulombic forces.

■ Solution:

Since $Q_1 = Q_3 = 10 \text{ nC}$ and $R_{12} = R_{32}$, we have $F_{12} = F_{32}$ (the same magnitude).

$$\vec{F}_{12} = 4.0(\hat{y} - \hat{z}) \text{ nN}$$

$$\vec{F}_{32} = 4.0(\hat{y} + \hat{z}) \text{ nN}$$

$$\vec{F} = \vec{F}_{12} + \vec{F}_{32} = 8.0\hat{y} \text{ nN}$$



Coulomb's Law

In general

The total force experienced by charge Q_i :

$$\vec{F}_i = \sum_{j=1}^n \vec{F}_{ji} = \sum_{j=1}^n \frac{Q_i Q_j}{4\pi\epsilon_0 R_{ji}^2} \hat{R}_{ji}$$

Electric Field Intensity

- A fixed charge Q_1 at the origin exerts force to a test charge Q_2 placed at the position $\vec{R}$:

$$\vec{F}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R_{12}^2} \hat{a}_{12} = \frac{Q_1 Q_2}{4\pi\epsilon_0 R^2} \hat{R}$$

- If we move Q_2 around, we can calculate a field/distribution of the Coulomb force at every point in space.



Coulomb's Law

Electric Field Intensity

- Define: **Electric Field Intensity** $\vec{E}_1$ from charge Q_1 :

$$\vec{E}_1 = \frac{\vec{F}_{12}}{Q_2} = \frac{Q_1}{4\pi\epsilon_0 R_{12}^2} \hat{R}_{12}$$

- Coulomb's law can be rewritten as:

$$\vec{E} = \frac{Q}{4\pi\epsilon_0 R^2} \hat{R}$$

This is the physical nature of charge Q .

- More explicitly, the electric field intensity can be calculated as:

$$\vec{E}(\vec{r}) = \frac{Q(\vec{r}_o)}{4\pi\epsilon_0 |\vec{r} - \vec{r}_o|^2} \hat{R}$$

where $\hat{R} = \frac{\vec{r} - \vec{r}_o}{|\vec{r} - \vec{r}_o|}$.

— Field produced at $\vec{r}$ by charge Q located at $\vec{r}_o$



HOWARD
UNIVERSITY

Coulomb's Law

Electric Field Intensity

- Total electric field produced by multiple charges:

$$\vec{E}(\vec{r}) = \sum_{j=1}^n \frac{Q_j(\vec{r}_j)}{4\pi\epsilon_0 |\vec{r} - \vec{r}_j|^2} \frac{\vec{r} - \vec{r}_j}{|\vec{r} - \vec{r}_j|}$$

- Units:

$$\begin{aligned} \vec{E} &= \frac{\vec{F}}{Q} \\ [\vec{E}] &= \frac{[\vec{F}]}{[Q]} \\ &= \frac{\text{N}}{\text{C}} = \frac{\text{J}}{\text{m} \cdot \text{V}} \\ &= \frac{\text{V}}{\text{m}} \end{aligned}$$



HOWARD
UNIVERSITY

Coulomb's Law

Example 2.4

Find the electric field intensity in Examples 2.2 and 2.3.

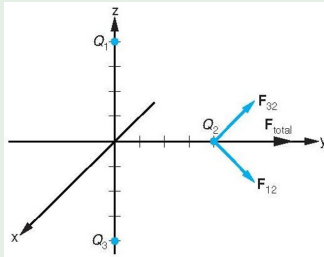


Figure: Example 2.3.

■ Solution:

Test charge $Q_2 = 2 \text{ nC} = 2 \times 10^{-9} \text{ C}$ in both examples.

In Example 2.2:

$$\vec{F}_{12} = 4.0(\hat{y} - \hat{z}) \times 10^{-9} \text{ N}$$

$$\vec{E}_1 = \frac{\vec{F}_{12}}{Q_2} = 2.0(\hat{y} - \hat{z}) \text{ V/m}$$

In Example 2.3:

$$\vec{F}_{\text{tot}} = 8.0\hat{y} \text{ nN}$$

$$\vec{E}_{\text{tot}} = \frac{\vec{F}_{\text{tot}}}{Q_2} = \frac{8.0\hat{y} \text{ nN}}{2.0 \text{ nN}} = 4.0\hat{y} \text{ V/m}$$



HOWARD
UNIVERSITY

Coulomb's Law

Field Lines

- Visualize fields by using **field lines**.
- Field lines: Lines that follow the direction of field vectors at convenient point in space.

Field vectors surrounding a positive charge

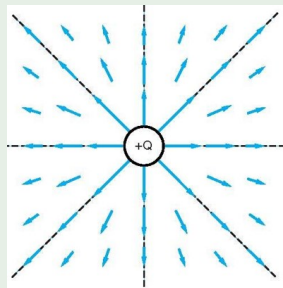


Figure: Field vectors plotted within a regular grid in two-dimensional space surrounding a point charge.

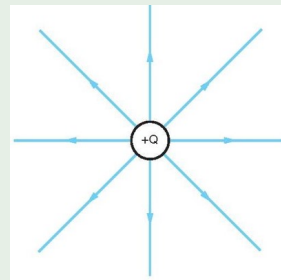


Figure: Some of these field vectors can easily be joined by field lines that emanate from the positive charge.

- The density of field lines represents the field strength.



HOWARD
UNIVERSITY

Coulomb's Law

Field Lines

- Visualize fields by using **field lines**.
- Field lines: Lines that follow the direction of field vectors at convenient point in space.

Field lines for a pair of opposite charges

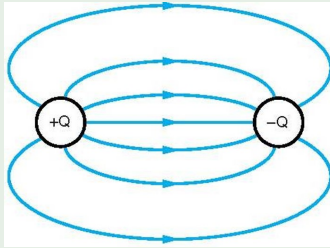


Figure: Field lines for a pair of opposite charges.

- Field is produced by a charge, regardless to the presence of another charge.
- Field lines visualize the vector field distribution:
 - Direction: lines with arrows;
 - Magnitude: density of the lines at a give location.



Spherical Coordinate System

A point charge located at the origin

Coulomb's law:

$$\begin{aligned}\vec{E}(\vec{r}) &= \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \\ \hat{r} &= \frac{\vec{r}}{r} = \frac{x\hat{x} + y\hat{y} + z\hat{z}}{\sqrt{x^2 + y^2 + z^2}} \\ \Rightarrow \vec{E}(\vec{r}) &= \frac{Q}{4\pi\epsilon_0 r^3} \vec{r} = \frac{Q}{4\pi\epsilon_0} \frac{x\hat{x} + y\hat{y} + z\hat{z}}{(x^2 + y^2 + z^2)^{3/2}}\end{aligned}$$

- Direction: $\hat{r}$
- Magnitude:

$$|\vec{E}(\vec{r})| = \frac{|Q|}{4\pi\epsilon_0 r^2} \propto \frac{1}{r^2}$$

- If expressed in the **Spherical Coordinate System** (SCS):

$$\vec{E}(\vec{r}) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$



Spherical Coordinate System

Spherical Coordinate System

A point in the SCS is represented by the orthogonal points (r, θ, ϕ) :

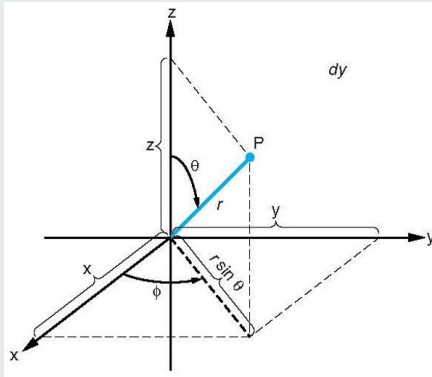


Figure: The spherical coordinate system is represented by the orthogonal points (r, θ, ϕ) .

- **r : Radial distance**, or range, from origin to point P :

$$0 \leq r < \infty \quad (\text{m})$$

- **θ : Colatitude**: Measured from $+z$ axis:

$$0 \leq \theta \leq \pi \quad (\text{radians, rad})$$

- **ϕ : Azimuthal angle**: Measured from $+x$ axis as move around the z axis, which is an angle in the xy plane:

$$0 \leq \phi \leq 2\pi \quad (\text{rad})$$



Spherical Coordinate System

Spherical Coordinate System

A point in the SCS is represented by the orthogonal points (r, θ, ϕ) :

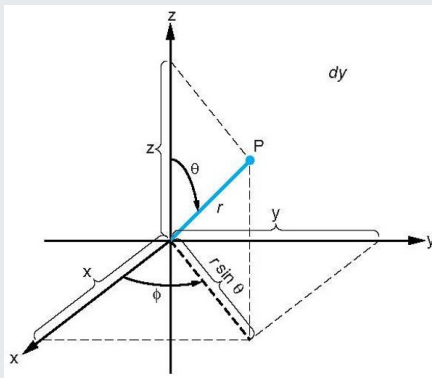


Figure: The spherical coordinate system is represented by the orthogonal points (r, θ, ϕ) .

Conversion of coordinates (Position vectors):

- $(x, y, z) \rightarrow (r, \theta, \phi)$:

$$r = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \cos^{-1} \left(\frac{z}{r} \right)$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

- $(r, \theta, \phi) \rightarrow (x, y, z)$:

$$x = r \sin \theta \cos \phi$$

$$y = r \sin \theta \sin \phi$$

$$z = r \cos \theta$$



Spherical Coordinate System

Spherical Coordinate System

Unit vectors in the SCS ($\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$):

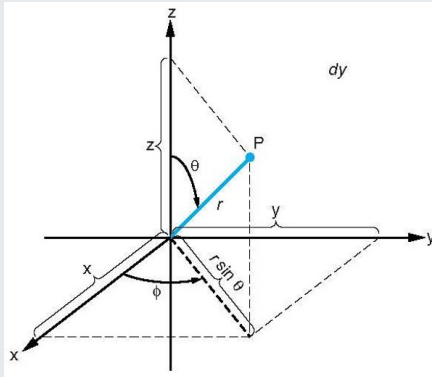


Figure: The spherical coordinate system is represented by the orthogonal points (r, θ, ϕ) .

- $\hat{r}$, $\hat{\theta}$, $\hat{\phi}$: Mutually orthogonal, right-hand system.
- Unit vectors $\hat{r}$, $\hat{\theta}$, and $\hat{\phi}$ depend on the location of $P = \vec{r} = (r, \theta, \phi)$, specifically, on θ and ϕ :

$$\hat{r} = \hat{r}(\theta, \phi)$$

$$\hat{\theta} = \hat{\theta}(\theta, \phi)$$

$$\hat{\phi} = \hat{\phi}(\phi)$$

In general:

	$\hat{x}$	$\hat{y}$	$\hat{z}$
$\hat{r}$	$\sin \theta \cos \phi$	$\sin \theta \sin \phi$	$\cos \theta$
$\hat{\theta}$	$\cos \theta \cos \phi$	$\cos \theta \sin \phi$	$-\sin \theta$
$\hat{\phi}$	$-\sin \phi$	$\cos \phi$	0

Table: Conversion of Unit Vectors



Spherical Coordinate System

Summation & Subtraction

$$\vec{A}(\vec{r}_1) = (A_r, A_\theta, A_\phi)$$

$$\vec{B}(\vec{r}_2) = (B_r, B_\theta, B_\phi)$$

$$\Rightarrow \vec{A}(\vec{r}_1) \pm \vec{B}(\vec{r}_2) = (A_r \pm B_r, A_\theta \pm B_\theta, A_\phi \pm B_\phi) \quad ?$$

- If $\hat{r}_1 = \hat{r}_2$, $\checkmark$;
- Otherwise, convert to Cartesian coordinates first, then sum up.



Spherical Coordinate System

Differential Elements

A differential element is defined by introducing a small increment at a point.

In spherical coordinate system: $\Delta r, \Delta \theta, \Delta \phi$.

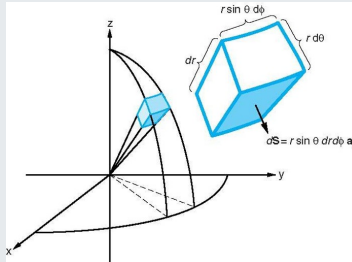


Figure: A differential element in the spherical coordinate system. One of the elements' six surfaces is shaded with the differential surface vector indicated.

- Differential lines: $dl = dr, r d\theta, \text{ or } r \sin \theta d\phi$ $d\vec{l} = \hat{r} dr, \hat{\theta} r d\theta, \text{ or } \hat{\phi} r \sin \theta d\phi$
- Differential surfaces: $dS = r dr d\theta, r \sin \theta dr d\phi, \text{ or } r^2 \sin \theta d\theta d\phi$
 $d\vec{S} = \pm \hat{\phi} r dr d\theta, \pm \hat{\theta} r \sin \theta dr d\phi, \text{ or } \pm \hat{r} r^2 \sin \theta d\theta d\phi$
- Differential volumes: $dV = r^2 \sin \theta dr d\theta d\phi$



Spherical Coordinate System

Example 2.5

Find the volume of a sphere with a radius R .

$$\begin{aligned}
 \iiint 1 dV &= \iiint r^2 \sin \theta dr d\theta d\phi \\
 &= \int_0^R r^2 dr \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \\
 &= \frac{1}{3} r^3 \Big|_0^R (-\cos \theta) \Big|_0^\pi 2\pi \\
 &= \frac{R^3}{3} \cdot 2 \cdot 2\pi \\
 &= \frac{4}{3} \pi R^3
 \end{aligned}$$

Find the surface area of a sphere with a radius R .

$$\begin{aligned}
 \iint 1 dS &= \iint r^2 \sin \theta d\theta d\phi \\
 &= R^2 \int_0^\pi \sin \theta d\theta \int_0^{2\pi} d\phi \\
 &= R^2 \cdot 2 \cdot 2\pi \\
 &= 4\pi R^2
 \end{aligned}$$



Cylindrical Coordinate System

Coordinate Systems

- Spherical symmetry $\rightarrow$ Spherical coordinate system
- Cylindrical symmetry $\rightarrow$ Cylindrical coordinate system
- Orthogonal components in cylindrical coordinate system:

$$(\rho, \phi, z)$$

- Point vector:

$$P = (x, y, z) = (\rho, \phi, z) = (r, \theta, \phi)$$

Cylindrical Coordinate System

Cylindrical Coordinate System (CCS)

A point in the CCS is represented by the orthogonal points (ρ, ϕ, z) :

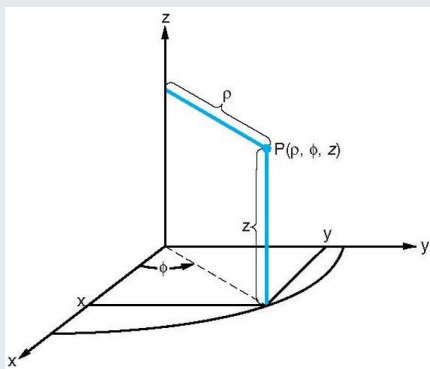


Figure: Cylindrical coordinate system.

- ρ : **Radial distance** from z axis to point P :

$$0 \leq \rho < \infty \quad (\text{m})$$

- ϕ : **Azimuthal angle**: Measured from $+x$ axis as move around the z axis; the same as defined in spherical coordinate system:

$$0 \leq \phi \leq 2\pi \quad (\text{rad})$$

- z : z coordinate; the same as defined in Cartesian coordinate system:

$$-\infty < z < \infty \quad (\text{m})$$

Cylindrical Coordinate System

Cylindrical Coordinate System

A point in the CCS is represented by the orthogonal points (ρ, ϕ, z) :

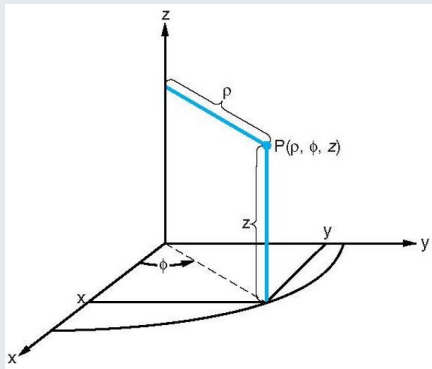


Figure: Cylindrical coordinate system.

Conversion of coordinates (Position vectors):

■ $(x, y, z) \rightarrow (\rho, \phi, z)$:

$$\rho = \sqrt{x^2 + y^2}$$

$$\phi = \tan^{-1} \left(\frac{y}{x} \right)$$

$$z = z$$

■ $(\rho, \phi, z) \rightarrow (x, y, z)$:

$$x = \rho \cos \phi$$

$$y = \rho \sin \phi$$

$$z = z$$



HOWARD
UNIVERSITY

Cylindrical Coordinate System

Cylindrical Coordinate System

Unit vectors in the CCS ($\hat{\rho}$, $\hat{\phi}$, and $\hat{z}$):

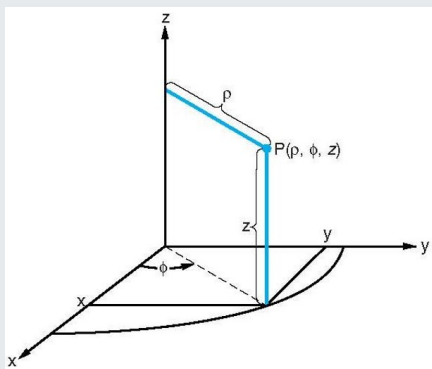


Figure: Cylindrical coordinate system.

■ $\hat{\rho}$, $\hat{\phi}$, and $\hat{z}$: Mutually orthogonal, right-hand system.

■ Unit vectors $\hat{\rho}$, and $\hat{\phi}$ depend on the location of $P = \vec{r} = (\rho, \phi, z)$, specifically, on ϕ :

$$\hat{\rho} = \hat{\rho}(\phi)$$

$$\hat{\phi} = \hat{\phi}(\phi)$$

In general:

	$\hat{x}$	$\hat{y}$	$\hat{z}$
$\hat{\rho}$	$\cos \phi$	$\sin \phi$	0
$\hat{\phi}$	$-\sin \phi$	$\cos \phi$	0
$\hat{z}$	0	0	1

Table: Conversion of Unit Vectors



HOWARD
UNIVERSITY

Cylindrical Coordinate System

Differential Elements

A differential element is defined by introducing a small increment at a point.

In cylindrical coordinate system: $\Delta\rho$, $\Delta\phi$, Δz .

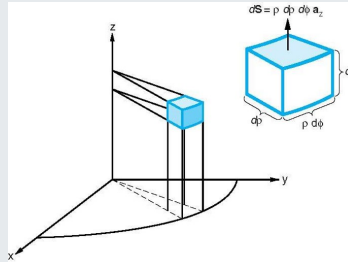


Figure: A differential element in cylindrical coordinates.

- Differential lines: $dl = d\rho, \rho d\phi, \text{ or } dz$ $d\vec{l} = \hat{\rho} d\rho, \hat{\phi} \rho d\phi, \text{ or } \hat{z} dz$
- Differential surfaces: $dS = \rho d\rho d\phi, d\rho dz, \text{ or } \rho d\phi dz$
 $d\vec{S} = \pm \hat{z} \rho d\rho d\phi, \pm \hat{\phi} d\rho dz, \text{ or } \pm \hat{\rho} \rho d\phi dz$
- Differential volumes: $dV = \rho d\rho d\phi dz$



Infinite Length Line of Charge

Line Charge

- Line charge:
 An amount of charge (in Coulombs) evenly distributed along every meter of a line.
- Line charge density: ρ_L (C/m);
 Surface charge density: ρ_S (C/m²);
 Volume charge density: ρ_V (C/m³);



Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

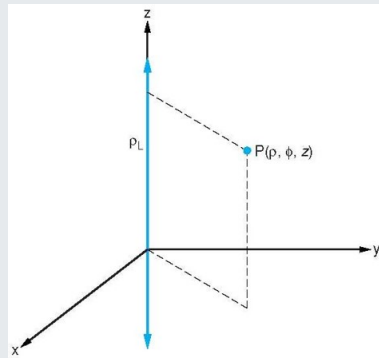


Figure: An infinite length line of charge, ρ_L , is placed along the z axis. We wish to find $\vec{E}$ at an arbitrary point $P(\rho, \phi, z)$.

Physical picture:

- In general: $\vec{E} = \vec{E}(\vec{r})$
- Location dependence: $\vec{r} = (\rho, \phi, z)$
- Infinitely long: Not a function of z
- Cylindrical symmetry: Not a function of ϕ
- $\rightarrow \vec{E}(\vec{r}) = \vec{E}(\rho)$

Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

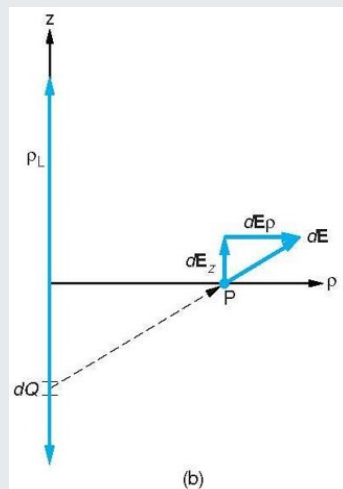


Figure: (b) Redrawn problem in the ρ - z plane with field components for a charge element dQ as indicated.

Field component:

- In general: $\vec{E} = (E_\rho, E_\phi, E_z)$
- In the ρ - z plane:
 - $d\vec{E}$ has only E_ρ and E_z components;

Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

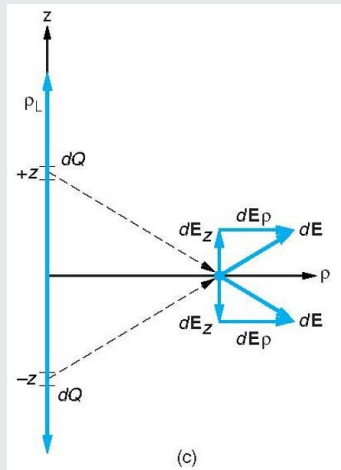


Figure: (c) Adding the field components for a second charge element dQ above the radial axis.

Field component:

- In general: $\vec{E} = (E_\rho, E_\phi, E_z)$
- In the ρ - z plane:
 $d\vec{E}$ has only E_ρ and E_z components;
 There is always dQ' that produces $d\vec{E}'$ to cancel the E_z component.
- $\rightarrow \vec{E}(\rho) = E_\rho(\rho) \hat{\rho}$: ρ component that is a function of ρ coordinate.



Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

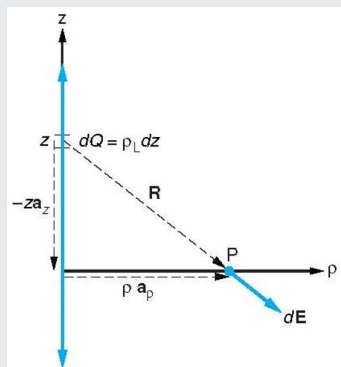


Figure: Sketch to show where all the components come from to set up the integral.

Mathematical calculation:

- 1 Field due to a point charge:

$$\vec{E}(\vec{r}) = \frac{Q}{4\pi\epsilon_0 R^2} \hat{R}$$

- 2 Along a piece of differential length dz , the total amount of charge:

$$dQ = \rho_L dz$$

- 3 Principle of superposition: Total field =
Summation/Integration of differential field:

$$\vec{E}(\vec{r}) = \int \frac{dQ}{4\pi\epsilon_0 R^2} \hat{R}$$

— The start point of calculation.



Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

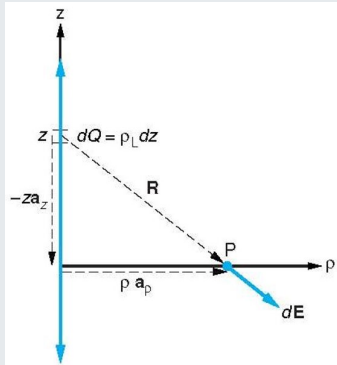


Figure: Sketch to show where all the components come from to set up the integral.

Mathematical calculation:

$$\vec{E}(\vec{r}) = \int \frac{dQ}{4\pi\epsilon_0 R^2} \hat{R}$$

$$\vec{R} = \rho\hat{\rho} - z\hat{z}$$

$$R = \sqrt{\rho^2 + z^2}$$

$$\hat{R} = \frac{\vec{R}}{R}$$

As a result:

$$\begin{aligned} \vec{E}(\vec{r}) &= \int_{-\infty}^{+\infty} \frac{\rho_L dz}{4\pi\epsilon_0 R^2} \frac{\vec{R}}{R} \\ &= \frac{\rho_L}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{\rho\hat{\rho} - z\hat{z}}{(\rho^2 + z^2)^{3/2}} dz \\ &= \frac{\rho_L}{4\pi\epsilon_0} \left[\int_{-\infty}^{+\infty} \frac{\rho\hat{\rho}}{(\rho^2 + z^2)^{3/2}} dz - \int_{-\infty}^{+\infty} \frac{z\hat{z}}{(\rho^2 + z^2)^{3/2}} dz \right] \end{aligned}$$



Infinite Length Line of Charge

Electrostatic field produced by an infinite length line of charge

Given line charge ρ_L placed along the z axis, find the electric field $\vec{E}(\vec{r})$ at location $\vec{r} = (\rho, \phi, z)$.

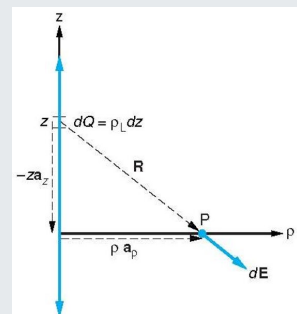


Figure: Sketch to show where all the components come from to set up the integral.

Mathematical calculation:

$$\vec{E}(\vec{r}) = \frac{\rho_L}{4\pi\epsilon_0} \left[\int_{-\infty}^{+\infty} \frac{\rho\hat{\rho}}{(\rho^2 + z^2)^{3/2}} dz - \int_{-\infty}^{+\infty} \frac{z\hat{z}}{(\rho^2 + z^2)^{3/2}} dz \right]$$

The second term in this expression is an odd function of z $\rightarrow$ vanishes upon integration along z , as long as the integration range is symmetric w.r.t. z .

$$\vec{E}(\vec{r}) = \frac{\rho_L \rho \hat{\rho}}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{dz}{(\rho^2 + z^2)^{3/2}}$$

By utilizing the following integral:

$$\int \frac{dx}{(x^2 + a^2)^{3/2}} = \frac{x}{a^2 \sqrt{x^2 + a^2}} = \frac{\text{sgn}(x)}{a^2 \sqrt{1 + (\frac{a}{x})^2}}$$

and substituting $dx \rightarrow dz$, $x \rightarrow z$, $a \rightarrow \rho$, we have:

$$\vec{E}(\vec{r}) = \frac{\rho_L \rho \hat{\rho}}{4\pi\epsilon_0} \frac{\text{sgn}(z)}{\rho^2 \sqrt{1 + (\frac{\rho}{z})^2}} \Big|_{z=-\infty}^{z=+\infty} = \frac{\rho_L}{2\pi\epsilon_0} \frac{\hat{\rho}}{\rho} = E_\rho(\rho) \hat{\rho}$$

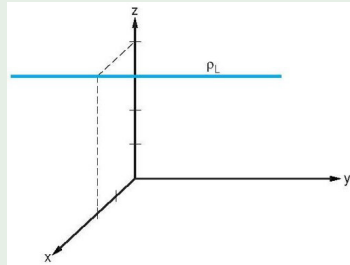


Infinite Length Line of Charge

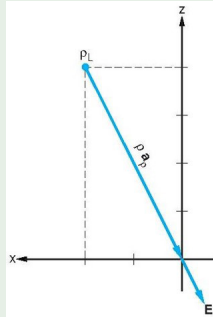
Example 2.7

An infinite line of charge $\rho_L = 4.0 \text{ nC/m}$ locates at $x = 2.0 \text{ m}$, $z = 4.0 \text{ m}$ along the y direction.

Find $\vec{E}(\vec{r} = 0)$.



(a)



(b)

$$\vec{E}(\vec{r}) = \frac{\rho_L}{2\pi\epsilon_0} \frac{\hat{\rho}}{\rho}$$

$$\vec{\rho} = \rho\hat{\rho} = -2\hat{x} - 4\hat{z}$$

$$\rho = \sqrt{2^2 + 4^2} = 2\sqrt{5} \text{ m}$$

$$\hat{\rho} = \frac{\vec{\rho}}{\rho} = -\frac{1}{\sqrt{5}}\hat{x} - \frac{2}{\sqrt{5}}\hat{z}$$

Figure: (a) An infinite length line charge is located at $x = 2$, $z = 4$. (b) Cross section at $y = 0$.

Note: $\hat{\rho}$ doesn't have to be the $\hat{\rho}$ in the cylindrical coordinate system. It's important to understand the physical meaning!

$$\vec{E}(\vec{r}) = \frac{4 \times 10^{-9}}{2\pi \times \frac{10^{-9}}{6\pi} \times 2\sqrt{5}} \left(-\frac{1}{\sqrt{5}}\hat{x} - \frac{2}{\sqrt{5}}\hat{z} \right) = -7.20\hat{x} - 14.4\hat{z} \text{ V/m}$$

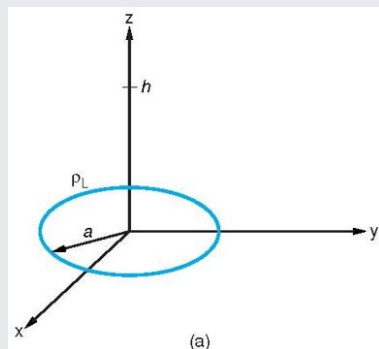


Ring of Charge

Electrostatic field produced by a ring of charge

A ring of charge, with the charge density ρ_L and a radius of a centered at the origin in the xy plane.

Find the electric field $\vec{E}$ along the z axis.



(a)

Figure: $\vec{E}$ is sought at point h on the z -axis from a ring of charge centered on the xy plane.

Physical picture:

- In general: $\vec{E} = \vec{E}(\vec{r})$
- Limit the interest on the z axis: $\vec{E} = \vec{E}(z)$
- Field component: $\vec{E} = (E_\rho, E_\phi, E_z)$



Ring of Charge

Electrostatic field produced by a ring of charge

A ring of charge, with the charge density ρ_L and a radius of a centered at the origin in the xy plane. Find the electric field $\vec{E}$ along the z axis.

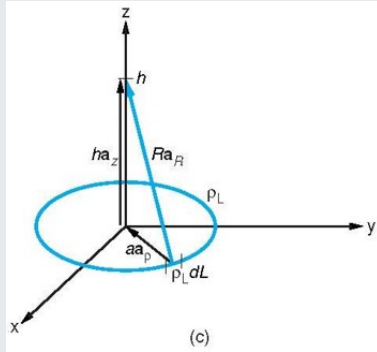


Figure: Terms for setting up the integral.

Mathematical calculation:

$$\vec{E}(\vec{r}) = \int \frac{dQ}{4\pi\epsilon_0 R^2} \hat{R}$$

$$\vec{R} = z\hat{z} - a\hat{\rho}$$

$$R = \sqrt{z^2 + a^2}$$

$$\hat{R} = \frac{\vec{R}}{R}$$

$$dQ = \rho_L dL = \rho_L a d\phi$$

As a result:

$$\vec{E}(\vec{r}) = \int_0^{2\pi} \frac{\rho_L a d\phi}{4\pi\epsilon_0 (z^2 + a^2)^{3/2}} (z\hat{z} - a\hat{\rho})$$



Ring of Charge

Electrostatic field produced by a ring of charge

A ring of charge, with the charge density ρ_L and a radius of a centered at the origin in the xy plane. Find the electric field $\vec{E}$ along the z axis.

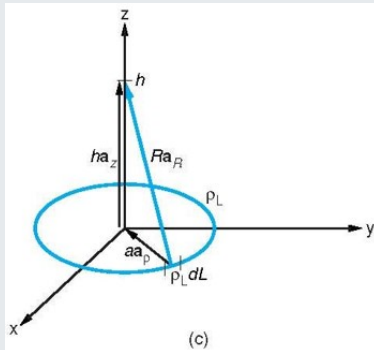


Figure: Terms for setting up the integral.

Mathematical calculation:

$$\begin{aligned} \vec{E}(\vec{r}) &= \int_0^{2\pi} \frac{\rho_L a d\phi}{4\pi\epsilon_0 (z^2 + a^2)^{3/2}} (z\hat{z} - a\hat{\rho}) \\ &= \frac{\rho_L a}{4\pi\epsilon_0 (z^2 + a^2)^{3/2}} \int_0^{2\pi} (z\hat{z} - a\hat{\rho}) d\phi \end{aligned}$$

Since: $\hat{z}$ is not a function of ϕ ;

$\hat{\rho}$ is a function of ϕ : $\hat{\rho} = \cos\phi\hat{x} + \sin\phi\hat{y}$

we have:

$$\begin{aligned} \vec{E}(\vec{r}) &= \frac{\rho_L a}{4\pi\epsilon_0 (z^2 + a^2)^{3/2}} \left[2\pi z\hat{z} - a \int_0^{2\pi} (\cos\phi\hat{x} + \sin\phi\hat{y}) d\phi \right] \\ &= \frac{\rho_L a z}{2\epsilon_0 (z^2 + a^2)^{3/2}} \hat{z} \end{aligned}$$

- If we are interested in calculating off-axis $\vec{E}$ field, numerical integration can be used.



Surface Charge

Surface Charge

- Surface charge:
 - A thin layer of charge distributed on a surface.
- For example: Charges distributed on a conducting surface/plane.
- Surface charge density: ρ_S (C/m²);



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

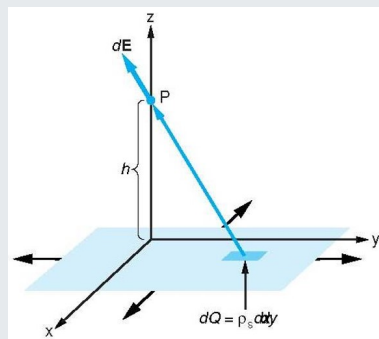


Figure: The point P is shown h above a current sheet of infinite extent occupying the xy plane.

Physical picture:

- In general: $\vec{E} = \vec{E}(\vec{r})$
- Due to symmetry (x and y extend infinitely):

$$\vec{E}(\vec{r}) = \vec{E}(z)$$

—> Doesn't change w.r.t. x or y .
- Field component: $\vec{E}(z) = E_z(z) \hat{z}$

—> Only has the z component due to cancellation.



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

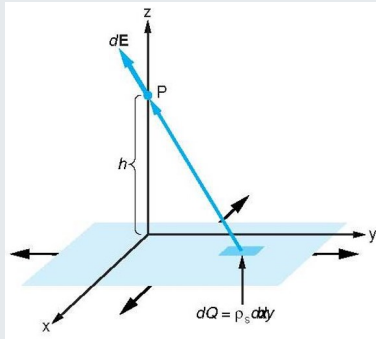


Figure: The point P is shown h above a current sheet of infinite extent occupying the xy plane.

Mathematical calculation:

- 1 Field due to a differential charge:

$$d\vec{E}(\vec{r}) = \frac{dQ}{4\pi\epsilon_0 R^2} \hat{R}$$

- 2 To calculate R , denote the source point:

$$\vec{r}' = (x', y', z') = (x', y', 0), \text{ where } x', y' \in (-\infty, +\infty)$$

and the field point:

$$\vec{r} = (x, y, z) = (0, 0, z), \text{ where } z \in (-\infty, +\infty)$$

we have

$$\vec{R} = \vec{r} - \vec{r}' = (x - x', y - y', z - z') = (-x', -y', z)$$

- 3 The differential charge dQ equals the charge density times the differential area:

$$dQ = \rho_S dS' = \rho_S dx' dy'$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

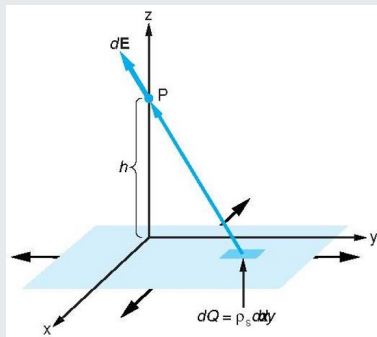


Figure: The point P is shown h above a current sheet of infinite extent occupying the xy plane.

Mathematical calculation:

Principle of superposition:

Total field = Integration of differential field

$$\vec{E}(\vec{r}) = \int d\vec{E}(\vec{r}) = \int \frac{dQ}{4\pi\epsilon_0 R^2} \hat{R}$$

$$\vec{R} = -x'\hat{x} - y'\hat{y} + z\hat{z}$$

$$R = \sqrt{x'^2 + y'^2 + z^2}$$

$$\hat{R} = \frac{\vec{R}}{R}$$

As a result:

$$\begin{aligned} \vec{E}(\vec{r}) &= \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{\rho_S dx' dy'}{4\pi\epsilon_0 R^2} \frac{\vec{R}}{R} \\ &= \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-x'\hat{x} - y'\hat{y} + z\hat{z}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' \end{aligned}$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

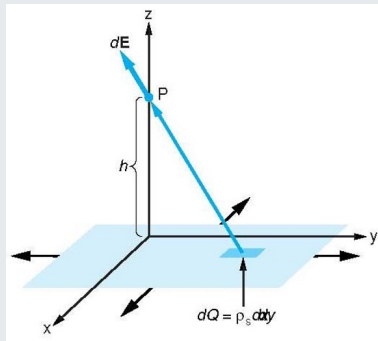


Figure: The point P is shown h above a current sheet of infinite extent occupying the xy plane.

Mathematical calculation:

$$\begin{aligned}\vec{E}(\vec{r}) &= \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-x'\hat{x} - y'\hat{y} + z\hat{z}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' \\ &= \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-x'\hat{x}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' \\ &\quad + \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-y'\hat{y}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' \\ &\quad + \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{z\hat{z}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy'\end{aligned}$$

- The integrations over x' and y' vanish due to symmetry:

$$\begin{aligned}\frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-x'\hat{x}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' &= 0 \\ \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{-y'\hat{y}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy' &= 0\end{aligned}$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

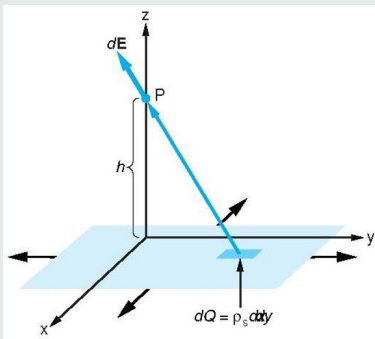


Figure: The point P is shown h above a current sheet of infinite extent occupying the xy plane.

Mathematical calculation:

$$\vec{E}(\vec{r}) = \frac{\rho_S}{4\pi\epsilon_0} \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} \frac{z\hat{z}}{(x'^2 + y'^2 + z^2)^{3/2}} dx' dy'$$

- Change of coordinate system:

$$\int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} f(x', y') dx' dy' = \int_0^{\infty} \int_0^{2\pi} f(\rho', \phi') \rho' d\phi' d\rho'$$

- we have:

$$\begin{aligned}\vec{E}(\vec{r}) &= \frac{\rho_S z \hat{z}}{4\pi\epsilon_0} \int_0^{\infty} \int_0^{2\pi} \frac{\rho' d\rho'}{(\rho'^2 + z^2)^{3/2}} d\phi' \\ &= \frac{\rho_S z \hat{z}}{4\pi\epsilon_0} \int_0^{2\pi} d\phi' \int_0^{\infty} \frac{\frac{1}{2} d\rho'^2}{(\rho'^2 + z^2)^{3/2}} \\ &= \frac{\rho_S z \hat{z}}{4\pi\epsilon_0} 2\pi \frac{1}{2} \int_{z^2}^{\infty} \frac{d(\rho'^2 + z^2)}{(\rho'^2 + z^2)^{3/2}} = \frac{\rho_S z \hat{z}}{4\epsilon_0} \int_{z^2}^{\infty} \tilde{\rho}^{-3/2} d\tilde{\rho} \\ &= \frac{\rho_S z \hat{z}}{4\epsilon_0} \left(-2\tilde{\rho}^{-1/2} \right) \Big|_{z^2}^{\infty} = \frac{\rho_S z \hat{z}}{4\epsilon_0} \left(2\frac{1}{z} \right) = \frac{\rho_S}{2\epsilon_0} \hat{z}\end{aligned}$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

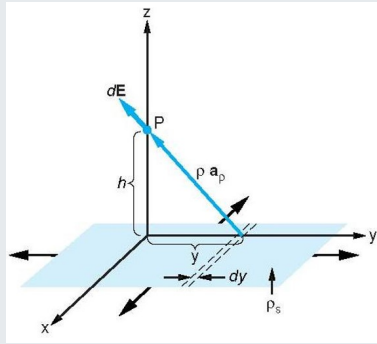


Figure: Infinite sheet of charge broken up into a continuous series of lines of charge.

An alternative approach:

Consider the surface charge as a continuous series of lines of charge.

- Field of line charge:

$$\vec{E}_L(\vec{r}) = \frac{\rho_L}{2\pi\epsilon_0\rho}\hat{\rho}$$

$$\rho_L = \rho_S dy'$$

- For the line charge located at y' and $z' = 0$:

$$\vec{\rho} = -y'\hat{y} + z\hat{z}$$

$$\rho = \sqrt{y'^2 + z^2}$$

$$\hat{\rho} = \frac{\vec{\rho}}{\rho}$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_S distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

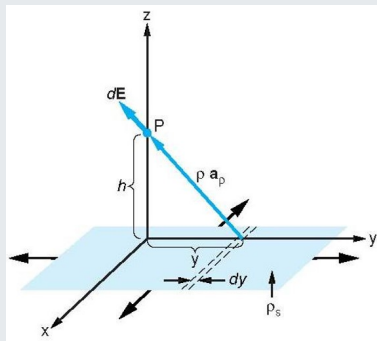


Figure: Infinite sheet of charge broken up into a continuous series of lines of charge.

An alternative approach:

- Field of surface charge:

$$\begin{aligned}\vec{E}(\vec{r}) &= \int_{-\infty}^{+\infty} \vec{E}_L(\vec{r}) = \int_{-\infty}^{+\infty} \frac{\rho_S}{2\pi\epsilon_0\rho}\hat{\rho}dy' \\ &= \frac{\rho_S}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{\vec{\rho}}{\rho^2} dy' \\ &= \frac{\rho_S}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{-y'\hat{y} + z\hat{z}}{y'^2 + z^2} dy' \\ &= \frac{\rho_S}{2\pi\epsilon_0} \int_{-\infty}^{+\infty} \frac{z\hat{z}}{y'^2 + z^2} dy'\end{aligned}$$

- Applying the following integration formula

$$\int_{-\infty}^{+\infty} \frac{dy'}{y'^2 + z^2} = \frac{1}{z} \tan^{-1} \left(\frac{y'}{z} \right) \Big|_{-\infty}^{+\infty} = \frac{\pi}{z}$$

- yields:

$$\vec{E}(\vec{r}) = \frac{\rho_S z \hat{z}}{2\pi\epsilon_0} \frac{\pi}{z} = \frac{\rho_S}{2\epsilon_0} \hat{z}$$



A Charge Sheet of Infinite Area

Electrostatic field produced by a charge sheet of infinite area

Given surface charge ρ_s distributed on the xy plane, find the electric field $\vec{E}(\vec{r})$ along the z axis.

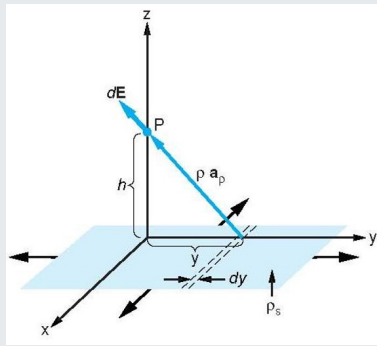


Figure: Infinite sheet of charge broken up into a continuous series of lines of charge.

- In general:

$$\vec{E}(\vec{r}) = \frac{\rho_s}{2\epsilon_0} \hat{n}$$

where $\hat{n}$ is the unit normal vector to the surface of the charge.

- Note:** The electric field $\vec{E}(\vec{r})$ is a constant vector, not a function of z (the distance from an observation point to the surface charge).
- Why?** The surface charge is infinite in size, which makes no difference in its field with respect to the distance of observation.



A Charge Sheet of Infinite Area

Example 2.8

An infinite surface of charge with $\rho_s = 10.0 \text{ nC/m}^2$ is located at $y = -2.0 \text{ m}$. Find the electric field at $\vec{r} = (0, 2, 1) \text{ m}$.

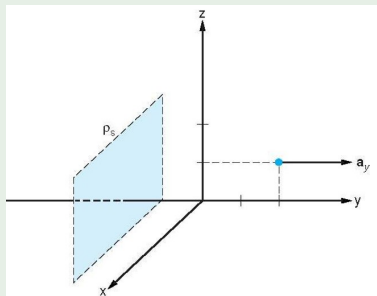


Figure: Infinite extent sheet of charge at $y = 2 \text{ m}$.

Solution:

- The electric field generated by an infinite surface of charge is a constant vector:

$$\vec{E}(\vec{r}) = \frac{\rho_s}{2\epsilon_0} \hat{n}$$

where $\hat{n} = \hat{y}$ is the unit normal vector to the surface of the charge that is pointing to the observation point.

- As a result

$$\begin{aligned} \vec{E}(\vec{r}) &= \frac{10.0 \times 10^{-9}}{2 \times \frac{10^{-9}}{36\pi}} \hat{y} \\ &= 565 \hat{y} \text{ V/m} \end{aligned}$$



Volume Charge

Volume Charge

- Volume charge:

Evenly distributed over a volume.

Volume charge density: ρ_V (C/m³)

- For example:

Plasma: ionized/charged particles (electrons/ions);

Doped semiconductor: PN junction, transistors: BJT, CMOS.

- Total charge:

$$Q = \int \rho_V(\vec{r}') dV$$



Volume Charge

Example 2.9

A volume charge with density $\rho_V = k r^2$ distributed in a sphere of radius a . Find the total charge.

Solution:

- Total charge:

$$\begin{aligned} Q &= \int \rho_V(\vec{r}') dV \\ &= \int_0^a \int_0^{2\pi} \int_0^\pi k r^2 r^2 \sin \theta d\theta d\phi dr \\ &= k \int_0^a r^4 dr \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \\ &= k \frac{r^5}{5} \Big|_0^a 2\pi (-\cos \theta) \Big|_0^\pi = \frac{4\pi}{5} k a^5 \end{aligned}$$

Electric field produced by a volume charge

$$\vec{E}(\vec{r}) = \int \frac{\rho_V(\vec{r}') \hat{R}}{4\pi\epsilon_0 R^2} dV = \int \frac{\rho_V(\vec{r}') (\vec{r} - \vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|^3} dV$$



Electric Flux

Electric Flux

- Notation: ψ (psi), in units of Coulombs (C).
- Flux lines originated at $+Q$ charges and terminated at $-Q$ charges.

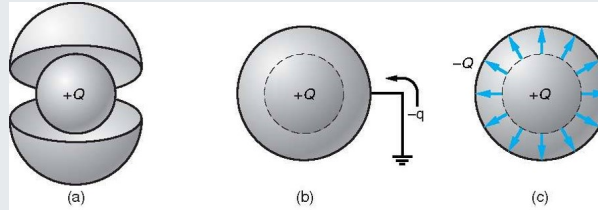


Figure: (a) Metallic sphere of radius a with charge $+Q$ is enclosed by a pair of hemispherical shells of radius b . (b) The outer shell is briefly grounded, (c) allowing a charge $-Q$ to accumulate on the outer shell.

- In the above example: $\psi = Q$ (C)
- **Electric flux density:** $\vec{D}$: Electric flux per unit area (C/m^2)

$$\vec{D} = \frac{\psi}{S} \hat{a} = \frac{\psi}{4\pi r^2} \hat{r}$$

- Note: at $\vec{r}$, $\vec{E} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$, we have the relation between $\vec{D}$ and $\vec{E}$ in free space (air, vacuum):

$$\vec{D} = \epsilon_0 \vec{E}$$



Electric Flux Density

Why define $\vec{D}$?

Electric flux density $\vec{D}$ relates to the number of flux lines emanating from one set of charges and terminating on the other, **independent of the media**.

$$\vec{D} = \epsilon \vec{E}$$

Here, ϵ denotes permittivity of the medium.

- The electric field intensity $\vec{E}$ is related to the material (medium) property.
- The amount of flux passing through a surface:

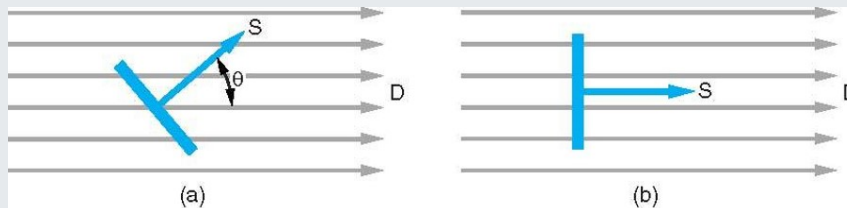


Figure: The flux through a surface that is at an angle to the direction of flux (a) is less than the flux through an equivalent surface normal to the direction of flux (b).

$$\psi = |\vec{D}| |\vec{S}| \cos \theta = \vec{D} \cdot \vec{S}$$



Dot Product

Dot Product (Inner Product)

- Dot product:

$$\vec{A} \cdot \vec{B} = |\vec{A}| |\vec{B}| \cos \theta_{AB}$$

where θ_{AB} denotes the angle between vectors $\vec{A}$ and $\vec{B}$.

- To calculate the inner product:

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (A_x \hat{x} + A_y \hat{y} + A_z \hat{z}) \cdot (B_x \hat{x} + B_y \hat{y} + B_z \hat{z}) \\ &= A_x B_x \hat{x} \cdot \hat{x} + A_x B_y \hat{x} \cdot \hat{y} + A_x B_z \hat{x} \cdot \hat{z} \\ &\quad + A_y B_x \hat{y} \cdot \hat{x} + A_y B_y \hat{y} \cdot \hat{y} + A_y B_z \hat{y} \cdot \hat{z} \\ &\quad + A_z B_x \hat{z} \cdot \hat{x} + A_z B_y \hat{z} \cdot \hat{y} + A_z B_z \hat{z} \cdot \hat{z} \\ &= A_x B_x + A_y B_y + A_z B_z \end{aligned}$$

- In different coordinate systems:

$$\vec{A} \cdot \vec{B} = A_x B_x + A_y B_y + A_z B_z = A_r B_r + A_\theta B_\theta + A_\phi B_\phi = A_\rho B_\rho + A_\phi B_\phi + A_z B_z$$

- Note: in spherical and cylindrical coordinate systems, the above expressions are true only when the respective unit vectors are the same.



Dot Product

Example 2.10

$\vec{A} = 5\hat{x} + 3\hat{z}$ and $\vec{B} = 2\hat{y} - 4\hat{z}$. Find $\vec{A} \cdot \vec{B}$.

Solution:

$$\begin{aligned} \vec{A} \cdot \vec{B} &= (5\hat{x} + 3\hat{z}) \cdot (2\hat{y} - 4\hat{z}) \\ &= 5 \cdot 0 + 0 \cdot 2 + 3 \cdot (-4) \\ &= -12 \end{aligned}$$

Dot Product

$\vec{A} \cdot \vec{B}$:

- 1 represent the amount of $\vec{A}$ that is in the direction of $\vec{B}$ ($\hat{B}$) multiplied by the magnitude of $\vec{B}$ ($\hat{B}$).
- 2 the projection of $\vec{A}$ onto $\vec{B}$.



Electric Flux

Calculation of the total electric flux

- If $\vec{D}$ is constant, S is a flat surface:

$$\psi = \vec{D} \cdot \vec{S}$$

- In general:

$$\psi = \int \vec{D} \cdot d\vec{S}$$

where $d\vec{S} = \hat{n} dS$.

Example 2.11

Given $\vec{D} = 3xy\hat{x} + 4x\hat{z}$ C/m² and the surface defined as $z = 0$, $x \in [0, 5]$ m, and $y \in [0, 3]$ m. Find the electric flux through the surface.

Solution:

$$d\vec{S} = \hat{z} dx dy$$

$$\begin{aligned} \psi &= \int \vec{D} \cdot d\vec{S} = \int_0^3 \int_0^5 4x dx dy \\ &= 4 \times \frac{1}{2} x^2 \Big|_0^5 \times y \Big|_0^3 = 150 \text{ (C)} \end{aligned}$$



Electric Flux

Example 2.12

Given $\vec{D} = 3r\hat{r} - 9r\hat{\theta} + 6\hat{\phi}$ C/m² and the spherical surface with a radius $r = 2$ m. Find the electric flux through the surface.

Solution:

$$\begin{aligned} \psi &= \int \vec{D} \cdot d\vec{S} \\ &= \int_0^{2\pi} \int_0^\pi (3r\hat{r} - 9r\hat{\theta} + 6\hat{\phi}) \cdot \hat{r} r^2 \sin \theta d\theta d\phi \\ &= 3r^3 \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta \\ &= 3r^3 \times 2\pi \times 2 \\ &= 96\pi \text{ (C)} \end{aligned}$$



Gauss's Law

Gauss's Law

The net electric flux through any closed surface is equal to the total charge enclosed by that surface.

- Mathematically:

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{enc}}$$

where $\oint$ denotes an integration performed over a closed surface.

—> **Integral form of Gauss's law.**

Application

Find the field distribution for problems with symmetry.

- Determine the symmetry, field dependence, and field components;
- Select an enclosing surface, **Gaussian surface**, with the normal direction either tangential to $\vec{D}$ ($\vec{D} \cdot d\vec{S} = 0$) or normal to $\vec{D}$ ($\vec{D} \cdot d\vec{S} = D dS$);

- Calculate the net electric flux:

$$\oint \vec{D} \cdot d\vec{S} = \int_{S_{\text{normal}}} D dS \quad (\text{if } \vec{D} \text{ is a constant on } S_{\text{normal}}) = D \int dS = D \cdot A$$

where A stands for the area of the surface.

- Calculate the electric flux density:

$$D = \frac{Q_{\text{enc}}}{\int dS} = \frac{Q_{\text{enc}}}{A}$$



Gauss's Law — Application

Example 2.13. Electrostatic field produced by a point charge at origin

Verify Coulomb's law by calculating the electrostatic field produced by a point charge at origin.

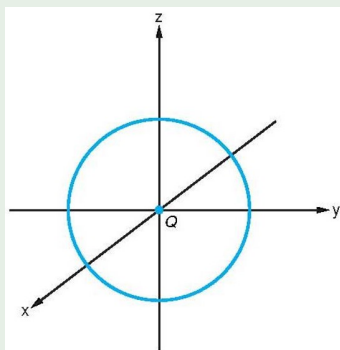


Figure: A spherical Gaussian surface surrounding a point charge at the origin.

Solution:

- Due to spherical symmetry:

$$\vec{D}(\vec{r}) = \hat{r} D_r(r)$$

- Choose the spherical surface at r as the Gaussian surface $d\vec{S} = r^2 \sin \theta d\theta d\phi \hat{r}$;

- Apply Gauss's law:

$$\begin{aligned} \oint \vec{D} \cdot d\vec{S} &= \iint D_r(r) \hat{r} \cdot \hat{r} r^2 \sin \theta d\theta d\phi \quad (D_r(r) \text{ is a constant at } r) \\ &= D_r(r) r^2 \int_0^{2\pi} d\phi \int_0^\pi \sin \theta d\theta = 4\pi r^2 D_r(r) = Q \end{aligned}$$

- This leads to

$$D_r(r) = \frac{Q}{4\pi r^2}$$

- Since $\vec{D} = \epsilon_0 \vec{E}$, we have

$$\vec{E}(\vec{r}) = \hat{r} E_r(r) = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$



Gauss's Law — Application

Example 2.14. Electrostatic field produced by an infinite length line of charge

Find the electrostatic field produced by an infinite length line of charge with density ρ_L along the z axis.

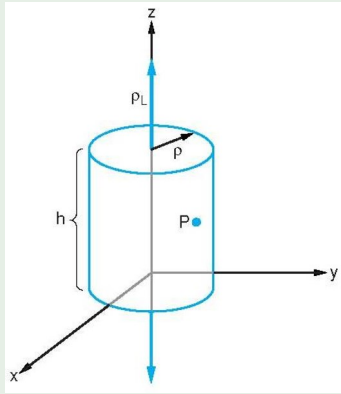


Figure: A Gaussian surface containing the point P is placed around a section of an infinite length line of charge density ρ_L occupying the z axis.

Solution:

- Due to cylindrical symmetry:

$$\vec{D}(\vec{r}) = \vec{D}(\rho, \phi, z) = \hat{\rho} D_\rho(\rho)$$

- Choose the Gaussian surface as the cylindrical surface at ρ with the top and bottom disks at $z = 0$ and h , respectively.

- Apply Gauss's law:

$$\begin{aligned} \oint \vec{D} \cdot d\vec{S} &= \int_{z=0,h} D_\rho(\rho) \hat{\rho} \cdot (\pm \hat{z}) \rho d\phi d\rho + \int_{\rho=\rho} D_\rho(\rho) \hat{\rho} \cdot \hat{\rho} \rho d\phi dz \\ &= D_\rho(\rho) \rho \int_0^h dz \int_0^{2\pi} d\phi = 2\pi h \rho D_\rho(\rho) = \rho_L h \end{aligned}$$

- This leads to

$$D_\rho(\rho) = \frac{\rho_L}{2\pi\rho}$$

- Since $\vec{D} = \epsilon_0 \vec{E}$, we have

$$\vec{E}(\vec{r}) = \hat{\rho} E_\rho(\rho) = \frac{\rho_L}{2\pi\epsilon_0\rho} \hat{\rho}$$



Gauss's Law — Application

Example 2.15. Electrostatic field produced by an infinite surface charge

Find the electrostatic field produced by an infinite-sized surface charge with density ρ_S at $z = 0$.

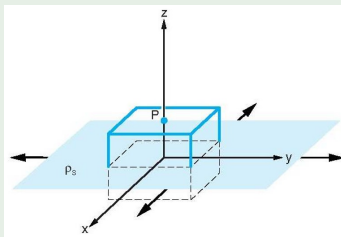


Figure: A rectangular box is employed as the Gaussian surface surrounding a section of sheet charge to determine the field resulting from an infinite extent sheet of charge density ρ_S .

Solution:

- Symmetric w.r.t. the z axis:

$$\vec{D}(\vec{r}) = \pm \hat{z} D_z(z)$$

- Choose the Gaussian surface as a cubical box bounded by $x \in [-x_0, x_0]$, $y \in [-y_0, y_0]$, and $z \in [-z_0, z_0]$.

- Apply Gauss's law:

$$\begin{aligned} \oint \vec{D} \cdot d\vec{S} &= \int_{\text{left \& right}} \pm \hat{z} D_z(z) \cdot (\pm \hat{y}) dy dz \\ &\quad + \int_{\text{front \& back}} \pm \hat{z} D_z(z) \cdot (\pm \hat{x}) dx dz \\ &\quad + \int_{\text{top \& bottom}} \pm \hat{z} D_z(z) \cdot (\pm \hat{z}) dx dy \\ &= \int_{z=z_0} \hat{z} D_z(z) \cdot \hat{z} dx dy + \int_{z=-z_0} (-\hat{z}) D_z(z) \cdot (-\hat{z}) dx dy \\ &= D_z 4x_0 y_0 + D_z 4x_0 y_0 = 8x_0 y_0 D_z \\ &= Q_{\text{enc}} = 4x_0 y_0 \rho_S \end{aligned}$$



Gauss's Law — Application

Example 2.15. Electrostatic field produced by an infinite surface charge

Find the electrostatic field produced by an infinite-sized surface charge with density ρ_S at $z = 0$.

Solution:

- Apply Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = 8x_0 y_0 D_z = Q_{\text{enc}} = 4x_0 y_0 \rho_S$$

- This leads to

$$D_z = \frac{\rho_S}{2} \rightarrow E_z = \frac{\rho_S}{2\epsilon_0}$$

- In general:

$$\vec{D} = \frac{\rho_S}{2} \hat{n} \quad \vec{E} = \frac{\rho_S}{2\epsilon_0} \hat{n}$$

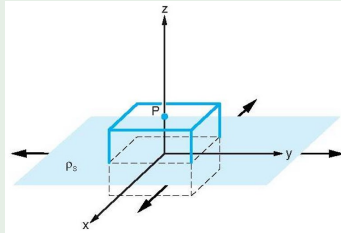


Figure: A rectangular box is employed as the Gaussian surface surrounding a section of sheet charge to determine the field resulting from an infinite extent sheet of charge density ρ_S .

Coaxial Cable

Electrostatic field in a coaxial cable

An infinitely long coaxial cable with an inner conductor of radius a , surrounded by a thin conductive shell at radius b . Assume

- 1 uniformly distributed volume charge with density ρ_V ; and
- 2 outer conductor being grounded.

Find the field distribution everywhere.

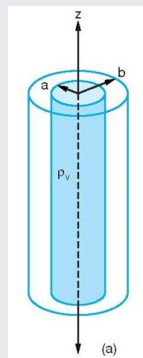


Figure: Coaxial cable with a solid inner conductor of radius a and charge density ρ_V surrounded by a grounded thin conductive shell at radius b .

Physical Analysis:

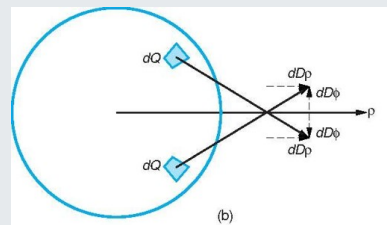


Figure: Cross section of the inner conductor with field components for a pair of differential charge elements.

- Due to axisymmetry and field cancellation:

$$\vec{D}(\vec{r}) = \hat{\rho} D_\rho(\rho)$$

Coaxial Cable

Electrostatic field in a coaxial cable

Find the field distribution everywhere in the coaxial cable.

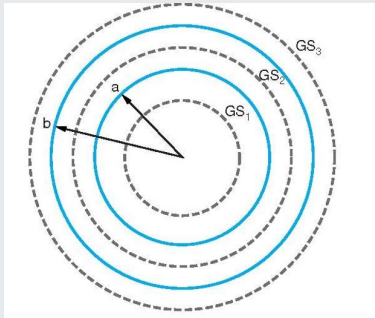


Figure: Cross section of the coaxial cable showing the three Gaussian surfaces.

Physical Analysis:

- Due to axisymmetry and field cancellation:

$$\vec{D}(\vec{r}) = \hat{\rho} D_{\rho}(\rho)$$

- The solution domain of interest can be decomposed into three regions:

- 1 $\rho \leq a$;
- 2 $a \leq \rho \leq b$;
- 3 $\rho \geq b$;

- Gaussian surfaces can be chosen as the cylindrical surfaces with $z \in [0, h]$.

- Apply Gauss's law:

$$\begin{aligned} \oint \vec{D} \cdot d\vec{S} &= \int_{\text{top \& bottom}} \hat{\rho} D_{\rho}(\rho) \cdot (\pm) \hat{z} dS \\ &+ \int_0^h \int_0^{2\pi} \hat{\rho} D_{\rho}(\rho) \cdot \hat{\rho} \rho d\phi dz \\ &= 2\pi h \rho D_{\rho} \end{aligned}$$



Coaxial Cable

Electrostatic field in a coaxial cable

Find the field distribution everywhere in the coaxial cable.

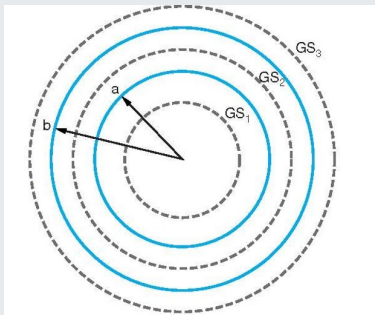


Figure: Cross section of the coaxial cable showing the three Gaussian surfaces.

Apply Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = 2\pi h \rho D_{\rho}$$

- (1) $\rho \leq a$:

$$\begin{aligned} Q_{\text{enc}} &= \int \rho_V dV = \int_0^h \int_0^{2\pi} \int_0^{\rho} \rho_V \rho d\phi d\rho dz \\ &= \rho_V \frac{1}{2} \rho^2 \Big|_0^{\rho} 2\pi h = \pi \rho^2 h \rho_V \end{aligned}$$

Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{enc}}$$

yields

$$2\pi h \rho D_{\rho} = \pi \rho^2 h \rho_V$$

which leads to

$$D_{\rho} = \frac{1}{2} \rho_V \rho$$



Coaxial Cable

Electrostatic field in a coaxial cable

Find the field distribution everywhere in the coaxial cable.

Apply Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = 2\pi h \rho D_\rho$$

(2) $a \leq \rho \leq b$:

$$\begin{aligned} Q_{\text{enc}} &= \int \rho_V dV = \int_0^h \int_0^a \int_0^{2\pi} \rho_V \rho d\phi d\rho dz \\ &= \rho_V \frac{1}{2} \rho^2 \bigg|_0^a 2\pi h = \pi a^2 h \rho_V \end{aligned}$$

Gauss's law leads to

$$D_\rho = \frac{a^2}{2\rho} \rho_V$$

(3) $\rho \geq b$:Outer surface is grounded, so charge on $\rho = b$ is $-Q$. We have:

$$Q_{\text{enc}} = \int \rho_V dV = 0$$

As a result:

$$D_\rho = 0 \quad (\rho \geq b)$$

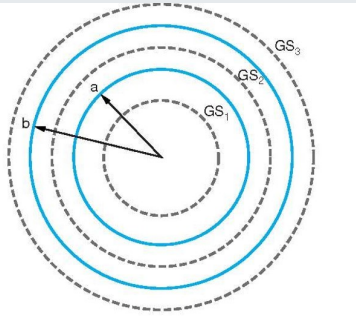


Figure: Cross section of the coaxial cable showing the three Gaussian surfaces.



Coaxial Cable

Electrostatic field in a coaxial cable

Find the field distribution everywhere in the coaxial cable.

Electrostatic field:

$$D_\rho = \begin{cases} \frac{\rho_V}{2} \rho, & \rho \leq a \\ \frac{a^2 \rho_V}{2} \frac{1}{\rho}, & a \leq \rho \leq b \\ 0, & \rho \geq b \end{cases}$$

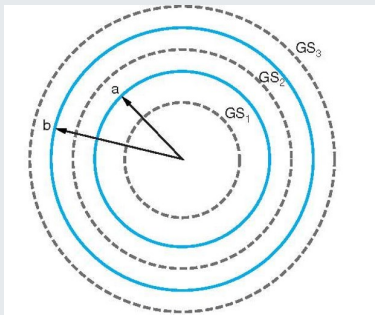
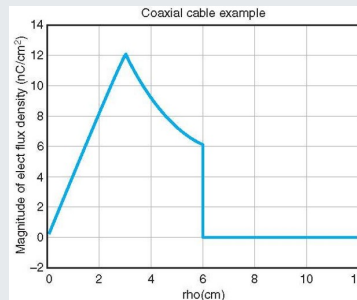


Figure: Cross section of the coaxial cable showing the three Gaussian surfaces.

Figure: D_ρ versus radius for the coaxial cable example.

Point Form of Gauss's Law

What does Gauss's law look like when applied to a point?

Apply Gauss's law to a point:

Apply Gauss's law to a small cubic element centered at $P = (x_0, y_0, z_0)$ and let the volume approach zero:

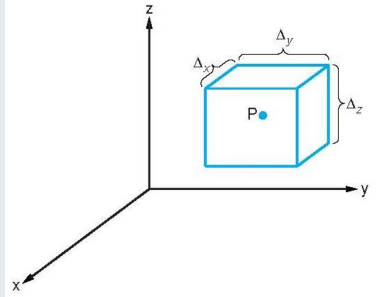


Figure: A differential volume element used to derive divergence.

Assume that we know:

$$\vec{D}(P) = D_x(x_0, y_0, z_0) \hat{x} + D_y(x_0, y_0, z_0) \hat{y} + D_z(x_0, y_0, z_0) \hat{z}$$

Apply Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = \int_{\text{front}} + \int_{\text{back}} + \int_{\text{left}} + \int_{\text{right}} + \int_{\text{top}} + \int_{\text{bottom}} = Q$$



Point Form of Gauss's Law

What does Gauss's law look like when applied to a point?

Apply Gauss's law to a point:

Apply Gauss's law to a small cubic element centered at $P = (x_0, y_0, z_0)$ and let the volume approach zero:

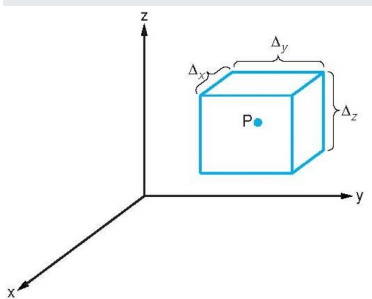


Figure: A differential volume element used to derive divergence.

Next, examine the integral on each surface:

(1) On the front surface:

$$\begin{aligned} \int_{\text{front}} \vec{D} \cdot d\vec{S} &= \int_{y_0 - \Delta y/2}^{y_0 + \Delta y/2} \int_{z_0 - \Delta z/2}^{z_0 + \Delta z/2} \vec{D}(x_0 + \frac{\Delta x}{2}, y, z) \cdot \hat{x} \, dy \, dz \\ &= D_x(x_0 + \frac{\Delta x}{2}) \Delta y \Delta z \end{aligned}$$

(2) On the back surface:

$$\begin{aligned} \int_{\text{back}} \vec{D} \cdot d\vec{S} &= \int_{y_0 - \Delta y/2}^{y_0 + \Delta y/2} \int_{z_0 - \Delta z/2}^{z_0 + \Delta z/2} \vec{D}(x_0 - \frac{\Delta x}{2}, y, z) \cdot (-\hat{x}) \, dy \, dz \\ &= -D_x(x_0 - \frac{\Delta x}{2}) \Delta y \Delta z \end{aligned}$$

(3) Combine these two terms:

$$\int_{\text{front}} + \int_{\text{back}} = \left[D_x(x_0 + \frac{\Delta x}{2}) - D_x(x_0 - \frac{\Delta x}{2}) \right] \Delta y \Delta z$$



Point Form of Gauss's Law

What does Gauss's law look like when applied to a point?

Apply Gauss's law to a point:

Apply Gauss's law to a small cubic element centered at $P = (x_0, y_0, z_0)$ and let the volume approach zero:

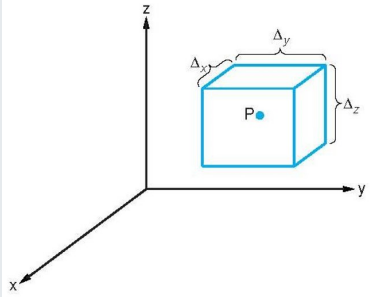


Figure: A differential volume element used to derive divergence.

Next, examine the integral on each surface:

(4) Similarly:

$$\int_{\text{right}} + \int_{\text{left}} = \left[D_y(y_0 + \frac{\Delta y}{2}) - D_y(y_0 - \frac{\Delta y}{2}) \right] \Delta x \Delta z$$

$$\int_{\text{top}} + \int_{\text{bottom}} = \left[D_z(z_0 + \frac{\Delta z}{2}) - D_z(z_0 - \frac{\Delta z}{2}) \right] \Delta x \Delta y$$

(5) As a result:

$$\begin{aligned} \text{LHS} &= \oint \vec{D} \cdot d\vec{S} \\ &= \left[D_x(x_0 + \frac{\Delta x}{2}) - D_x(x_0 - \frac{\Delta x}{2}) \right] \Delta y \Delta z \\ &\quad + \left[D_y(y_0 + \frac{\Delta y}{2}) - D_y(y_0 - \frac{\Delta y}{2}) \right] \Delta x \Delta z \\ &\quad + \left[D_z(z_0 + \frac{\Delta z}{2}) - D_z(z_0 - \frac{\Delta z}{2}) \right] \Delta x \Delta y \\ &= Q = \text{RHS} \end{aligned}$$



Point Form of Gauss's Law

What does Gauss's law look like when applied to a point?

Apply Gauss's law to a point:

Apply Gauss's law to a small cubic element centered at $P = (x_0, y_0, z_0)$ and let the volume approach zero:

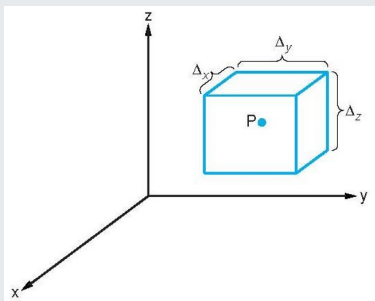


Figure: A differential volume element used to derive divergence.

Next, examine the integral on each surface:

(6) Divide both sides by $\Delta V = \Delta x \Delta y \Delta z$, and let

$\Delta V \rightarrow 0$:

$$\begin{aligned} \text{LHS} &= \lim_{\Delta V \rightarrow 0} \frac{\oint \vec{D} \cdot d\vec{S}}{\Delta V} \\ &= \lim_{\Delta x \rightarrow 0} \frac{D_x(x_0 + \frac{\Delta x}{2}) - D_x(x_0 - \frac{\Delta x}{2})}{\Delta x} \\ &\quad + \lim_{\Delta y \rightarrow 0} \frac{D_y(y_0 + \frac{\Delta y}{2}) - D_y(y_0 - \frac{\Delta y}{2})}{\Delta y} \\ &\quad + \lim_{\Delta z \rightarrow 0} \frac{D_z(z_0 + \frac{\Delta z}{2}) - D_z(z_0 - \frac{\Delta z}{2})}{\Delta z} \\ &= \lim_{\Delta V \rightarrow 0} \frac{Q}{\Delta V} = \rho_V \end{aligned}$$

(7) This yields:

$$\frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z} = \rho_V$$



Point Form of Gauss's Law

What does Gauss's law look like when applied to a point?

Apply Gauss's law to a point:

Apply Gauss's law to a small cubic element centered at $P = (x_0, y_0, z_0)$ and let the volume approach zero:

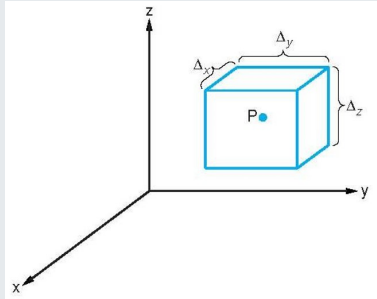


Figure: A differential volume element used to derive divergence.

Introducing **del operator**: ∇

- In Cartesian coordinate system:

$$\nabla = \frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z}$$

- The divergence of $\vec{D}$:

$$\nabla \cdot \vec{D} = \left(\frac{\partial}{\partial x} \hat{x} + \frac{\partial}{\partial y} \hat{y} + \frac{\partial}{\partial z} \hat{z} \right) \cdot (D_x \hat{x} + D_y \hat{y} + D_z \hat{z})$$

$$\boxed{\nabla \cdot \vec{D} = \rho_V}$$

→ **Gauss's law in point form (differential form).**

→ The amount of electric flux leaving the point.

- ρ_V : The divergence source of the electric flux density: $\rho_V > 0$: Flux source → diverge

$\rho_V < 0$: Flux sink → converge



HOWARD
UNIVERSITY

Point Form of Gauss's Law

Divergence Operation

- In Cartesian coordinate system:

$$\nabla \cdot \vec{D} = \frac{\partial D_x}{\partial x} + \frac{\partial D_y}{\partial y} + \frac{\partial D_z}{\partial z}$$

- In cylindrical coordinate system:

$$\nabla \cdot \vec{D} = \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z}$$

- In spherical coordinate system:

$$\nabla \cdot \vec{D} = \frac{1}{r^2} \frac{\partial r^2 D_r}{\partial r} + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (D_\theta \sin \theta) + \frac{1}{r \sin \theta} \frac{\partial D_\phi}{\partial \phi}$$

Divergence Theorem

- From Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = Q = \int \rho_V dV = \int \nabla \cdot \vec{D} dV$$

- we have

$$\boxed{\int \nabla \cdot \vec{D} dV = \oint \vec{D} \cdot d\vec{S}}$$

→ **Divergence theorem (Gauss theorem)**



HOWARD
UNIVERSITY

Point Form of Gauss's Law

Example 2.16

Given $\vec{D} = \rho^2 \hat{\rho}$, find the flux through the surface of a cylinder with $0 \leq z \leq h$, $\rho = a$ by evaluating:

(a) $\oint \vec{D} \cdot d\vec{S}$; and

(b) $\int \nabla \cdot \vec{D} dV$.

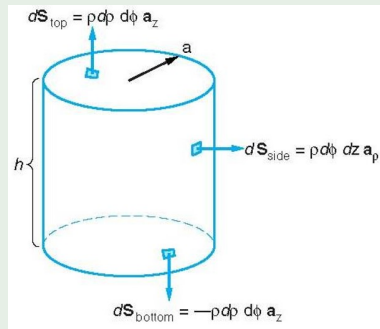


Figure: Cylindrical volume of Example 2.16 with differential surface vectors shown.

Solution:

(a)

$$\begin{aligned}
 \oint \vec{D} \cdot d\vec{S} &= \oint \rho^2 \hat{\rho} \cdot d\vec{S} \\
 &= \int_{\text{top}} \rho^2 \hat{\rho} \cdot \hat{z} dS + \int_{\text{bottom}} \rho^2 \hat{\rho} \cdot (-\hat{z}) dS \\
 &\quad + \int_{\text{side}} \rho^2 \hat{\rho} \cdot \hat{\rho} dS \\
 &= \int_0^h \int_0^{2\pi} \rho^2 \hat{\rho} \cdot \hat{\rho} d\phi dz \Big|_{\rho=a} \\
 &= a^3 \int_0^h \int_0^{2\pi} d\phi dz = 2\pi h a^3
 \end{aligned}$$



HOWARD
UNIVERSITY

Point Form of Gauss's Law

Example 2.16

Given $\vec{D} = \rho^2 \hat{\rho}$, find the flux through the surface of a cylinder with $0 \leq z \leq h$, $\rho = a$ by evaluating:

(a) $\oint \vec{D} \cdot d\vec{S}$; and

(b) $\int \nabla \cdot \vec{D} dV$.

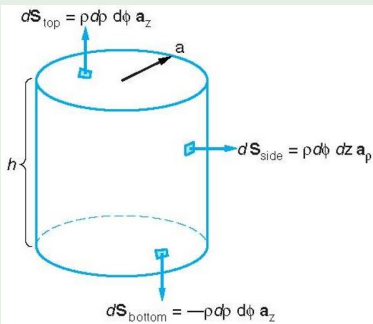


Figure: Cylindrical volume of Example 2.16 with differential surface vectors shown.

Solution:

(b) In cylindrical coordinate system:

$$\begin{aligned}
 \nabla \cdot \vec{D} &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho D_\rho) + \frac{1}{\rho} \frac{\partial D_\phi}{\partial \phi} + \frac{\partial D_z}{\partial z} \\
 &= \frac{1}{\rho} \frac{\partial}{\partial \rho} (\rho \rho^2) = \frac{1}{\rho} 3\rho^2 = 3\rho
 \end{aligned}$$

Hence,

$$\begin{aligned}
 \int \nabla \cdot \vec{D} dV &= \int 3\rho dV = \int_0^h \int_0^{2\pi} \int_0^a 3\rho \rho d\rho d\phi dz \\
 &= \rho^3 \Big|_0^a h 2\pi = 2\pi h a^3
 \end{aligned}$$



HOWARD
UNIVERSITY

Electric Potential

Electric Potential

- Electric potential: Voltage (V) in circuit analysis;
- How is it related to the electric field intensity $\vec{E}$?
- Work done by electrostatic force:

$$W = \int_a^b \vec{F} \cdot d\vec{l}$$

- Force exerted by the field: $\vec{F} = Q\vec{E}$, hence

$$W_E = Q \int_a^b \vec{E} \cdot d\vec{l}$$

This is the work done by the electric field.

- An external force moves the charge against the field, the work done is negative of the force W_E :

$$W_E = -Q \int_a^b \vec{E} \cdot d\vec{l}$$



Electric Potential

Electric Potential

- Define: **Electric potential difference**, V_{ba} :

The amount of work required to move a unit charge from point a to point b in an electric field:

$$\begin{aligned} V_{ba} &= \frac{W}{Q} = - \int_a^b \vec{E} \cdot d\vec{l} \\ &= V_b - V_a \text{ (V)} \end{aligned}$$

- In the above, V_a and V_b are referred to as **electrostatic potential** (absolute potential).
- The definition of an absolute potential requires a **reference**.
e.g.: A ground plane, or a plate, that is chosen (defined) as the zero potential reference.



Electric Potential

Electric Potential

- If a closed path is chosen, the integral is zero:

No potential difference between a point and itself.

$$\oint \vec{E} \cdot d\vec{l} = 0$$

—> Kirchhoff's voltage law.

This is only true for static field!



Electric Potential

Potential difference in space in the field of a point charge

- Electric field produced by a point charge at origin:

$$\vec{E}(\vec{r}) = E_r \hat{r} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

- Since electric field has only the E_r component, only motion in the radial direction will experience the potential change.

- Move from radius a to b :

$$\begin{aligned} V_{ba} &= - \int_a^b \vec{E} \cdot d\vec{l} = - \int_a^b \frac{Q}{4\pi\epsilon_0 r^2} \hat{r} \cdot \hat{r} dr \\ &= \frac{Q}{4\pi\epsilon_0} \left[\frac{1}{r} \right]_{r=a}^{r=b} = \frac{Q}{4\pi\epsilon_0} \left(\frac{1}{b} - \frac{1}{a} \right) \\ &= V_b - V_a \end{aligned}$$

- Choose $r = \infty$ as the reference point where $V_\infty = 0$ V,

$$V = \frac{Q}{4\pi\epsilon_0 r}$$

is the electrostatic potential produced by the point charge.

—> The work per coulomb required to pull a charge from ∞ to r .



Electric Potential

Potential produced by multiple charges

- For N discrete charges:

$$V(\vec{r}) = \sum_{i=1}^N \frac{Q_i}{4\pi\epsilon_0 |\vec{r} - \vec{r}_i|}$$

where $\vec{r}_i$ is the location of charge Q_i .

- For continuously distributed charges:

$$V(\vec{r}) = \int \frac{dQ(\vec{r}')}{4\pi\epsilon_0 |\vec{r} - \vec{r}'|}$$



Electric Potential

Gradient

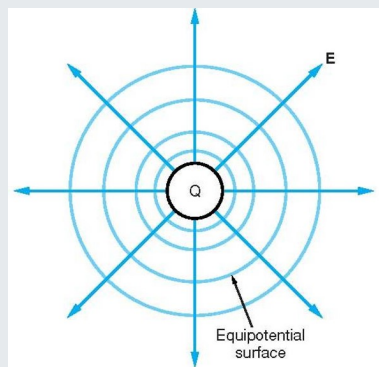


Figure: Equipotential lines are shown orthogonal to field lines for a point charge.

In the case of a point charge:

$$V = \frac{Q}{4\pi\epsilon_0 r} \rightarrow \vec{E} = -\nabla V = -\frac{\partial V}{\partial r} \hat{r} = \frac{Q}{4\pi\epsilon_0 r^2} \hat{r}$$

The electric field is in the direction where the potential decreases the fastest:

$$\vec{E}(\vec{r}) = -\nabla V(\vec{r})$$

—> **Gradient**

- In Cartesian coordinate system:

$$\nabla V = \frac{\partial V}{\partial x} \hat{x} + \frac{\partial V}{\partial y} \hat{y} + \frac{\partial V}{\partial z} \hat{z}$$

- In cylindrical coordinate system:

$$\nabla V = \frac{\partial V}{\partial \rho} \hat{\rho} + \frac{1}{\rho} \frac{\partial V}{\partial \phi} \hat{\phi} + \frac{\partial V}{\partial z} \hat{z}$$

- In spherical coordinate system:

$$\nabla V = \frac{\partial V}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{\phi}$$



Electric Potential

Three ways to find electric field distribution

- 1 Coulomb's law:

$$\vec{E}(\vec{r}) = \int \frac{dQ(\vec{r}')}{4\pi\epsilon_0 R^2} \hat{R}$$

- 2 Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = Q_{\text{enc}} \rightarrow \vec{E} = \frac{\vec{D}}{\epsilon_0}$$

- 3 Electrostatic potential:

$$V(\vec{r}) = \int \frac{dQ(\vec{r}')}{4\pi\epsilon_0 R} \rightarrow \vec{E} = -\nabla V$$

Why approach 3 is simpler than approach 1?

- $1/R$ vs. $1/R^2$ in the integration;
- scalar vs. vector integrand.



Electric Potential

Example 2.19

Surface charges with density ρ_S distribute on a circular disk with a radius of a . Find $\vec{E}$ field at $z = h$, on the z axis.

Solution:

- Electrostatic potential:

$$V(h) = \int \frac{dQ(\vec{r}')}{4\pi\epsilon_0 r}$$

$$dQ = \rho_S dS = \rho_S \rho d\phi d\rho$$

$$r = \sqrt{h^2 + \rho^2}$$

- we have:

$$\begin{aligned} V(h) &= \int_0^a \int_0^{2\pi} \frac{\rho_S \rho}{4\pi\epsilon_0 \sqrt{h^2 + \rho^2}} d\phi d\rho \\ &= \frac{\rho_S}{4\pi\epsilon_0} 2\pi \int_0^a \frac{\rho}{\sqrt{h^2 + \rho^2}} d\rho \\ &= \frac{\rho_S}{2\epsilon_0} \int_0^a \frac{\frac{1}{2}}{\sqrt{h^2 + \rho^2}} d\rho^2 \\ &= \frac{\rho_S}{4\epsilon_0} \int_0^a (h^2 + \rho^2)^{-\frac{1}{2}} d(h^2 + \rho^2) \end{aligned}$$

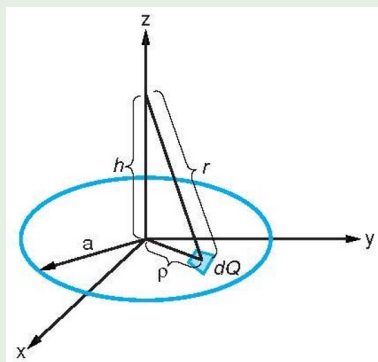


Figure: Disk of charge to find V and $\vec{E}$ on the z axis.



Electric Potential

Example 2.19

Surface charges with density ρ_S distribute on a circular disk with a radius of a . Find $\vec{E}$ field at $z = h$, on the z axis.

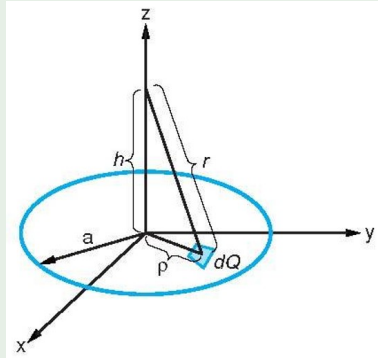


Figure: Disk of charge to find V and $\vec{E}$ on the z axis.

Solution:

Carrying out the remaining integration, with h denoting the z coordinate, yields

$$V(z) = \frac{\rho_S}{2\epsilon_0} \left(\sqrt{z^2 + a^2} - z \right)$$

Next, find $\vec{E}$ field:

$$\begin{aligned} \vec{E} &= -\nabla V(z) = -\frac{\partial V}{\partial z} \hat{z} \\ &= -\frac{\rho_S}{2\epsilon_0} \left(\frac{1}{2} \frac{2z}{\sqrt{z^2 + a^2}} - 1 \right) \hat{z} \\ &= \frac{\rho_S}{2\epsilon_0} \left(1 - \frac{z}{\sqrt{z^2 + a^2}} \right) \hat{z} \end{aligned}$$

Example: Practice

If $V = x^2yz$ V, find $\vec{E}(2, 3, 0)$.

Solution:



Electric Potential

Example 2.20

A pair of point charges of equal magnitude and opposite signs, $+Q$ and $-Q$, is located in very close proximity $d \rightarrow 0$. $\rightarrow$ **Electric dipole**, a useful model to describe the behavior of **dielectric materials**. Find the potential and field that's far away from the dipole.

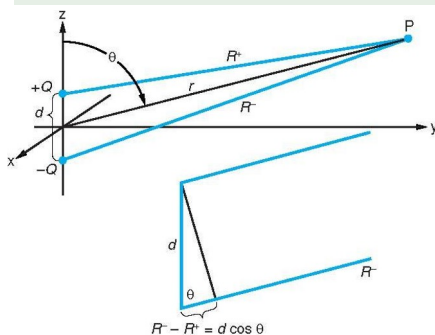


Figure: An electric dipole.

Solution:

- Far away: $r \gg d$
- Fields depend on r, θ , but not ϕ , due to symmetry.
- The total potential at P :

$$\begin{aligned} V &= \frac{+Q}{4\pi\epsilon_0 R^+} + \frac{-Q}{4\pi\epsilon_0 R^-} \\ &= \frac{Q}{4\pi\epsilon_0} \left(\frac{R^- - R^+}{R^+ R^-} \right) \end{aligned}$$

- Since $r \gg d$:

$$R^+ \approx R^- \approx r \Rightarrow R^+ R^- \approx r^2$$

- Since $r \gg d$:

$$R^+ // R^- \Rightarrow R^- - R^+ \approx d \cos \theta$$



Electric Potential

Example 2.20

A pair of point charges of equal magnitude and opposite signs, $+Q$ and $-Q$, is located in very close proximity $d \rightarrow 0$. $\rightarrow$ **Electric dipole**, a useful model to describe the behavior of **dielectric materials**. Find the potential and field that's far away from the dipole.

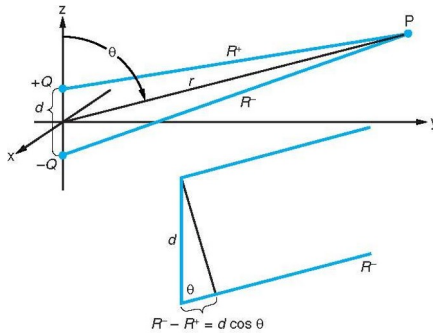


Figure: An electric dipole.

Solution:

■ As a result:

$$V = \frac{Q}{4\pi\epsilon_0} \left(\frac{R^- - R^+}{R^+ R^-} \right) = \frac{Qd \cos \theta}{4\pi\epsilon_0 r^2}$$

■ Electric field:

$$\begin{aligned} \vec{E} &= -\nabla V = -\frac{\partial V}{\partial r} \hat{r} - \frac{1}{r} \frac{\partial V}{\partial \theta} \hat{\theta} - \frac{1}{r \sin \theta} \frac{\partial V}{\partial \phi} \hat{\phi} \\ &= -\frac{Qd \cos \theta}{4\pi\epsilon_0} \frac{\partial}{\partial r} (r^{-2}) \hat{r} - \frac{1}{r} \frac{Qd}{4\pi\epsilon_0 r^2} \frac{\partial}{\partial \theta} (\cos \theta) \hat{\theta} \\ &= \frac{Qd \cos \theta}{2\pi\epsilon_0 r^3} \hat{r} + \frac{Qd \sin \theta}{4\pi\epsilon_0 r^3} \hat{\theta} \\ &= \frac{Qd}{4\pi\epsilon_0 r^3} (2 \cos \theta \hat{r} + \sin \theta \hat{\theta}) \end{aligned}$$



Materials

Material Property & Categorization

- The electromagnetic properties of material are described by **constitutive parameters**:
 - Permittivity (ϵ): describes dielectric materials, related to capacitance (C) in circuits;
 - Permeability (μ): associated with magnetic fields and inductance (L) in circuits;
 - Conductivity (σ): material's ability to conduct electricity, related to resistance (R) in circuits.
- **Homogeneous**: Properties are the same at every point;
 - Inhomogeneous**: $\epsilon(\vec{r})$, $\mu(\vec{r})$, and $\sigma(\vec{r})$;
- **Isotropic**: Properties are independent of direction ($\hat{x}$, $\hat{y}$, and $\hat{z}$);
 - Anisotropic**: Properties depend on direction ($\hat{x}$, $\hat{y}$, or $\hat{z}$), e.g., crystal;
- **Linear/Nonlinear**: Properties as a linear/nonlinear function of field.



Conductivity of Materials

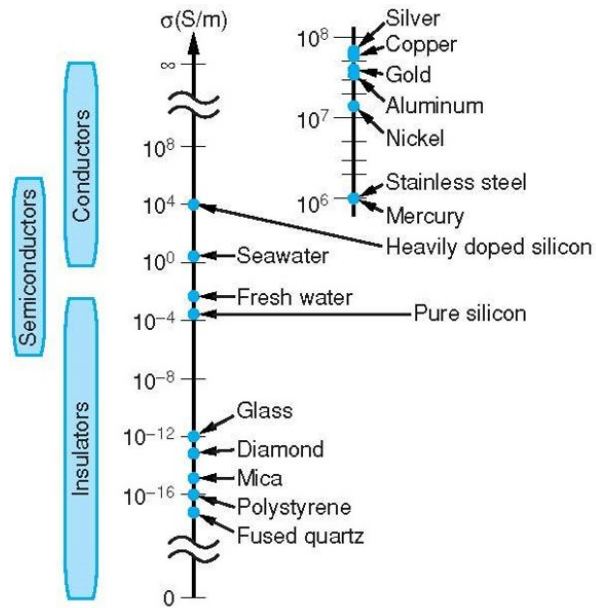


Figure: Conductivity chart (at room temperature).

Electric Current

Current Density

- Current density: $\vec{J}$ (A/m²)

Electric current:

$$I = \int \vec{J} \cdot d\vec{S}$$

Compare to electric flux:

$$\psi = \int \vec{D} \cdot d\vec{S}$$

- Types of current density:

Convection: Movement of charged particles through vacuum, air, or other non-conductive media, e.g., electron beams in a cathode ray tube (TV picture tube).

$$\vec{J} = \rho_V \vec{u}$$

with ρ_V the charge density and $\vec{u}$ the velocity.

Conduction: Movement of electrons through conductive media in response to an applied electric field, e.g., electricity in conducting wires.

$$\vec{J} = \sigma \vec{E}$$

→ Point form of Ohm's law.

Displacement: Displacement current density is a time-varying field component that allows current to flow between the plates of a capacitor.

$$\vec{J} = \frac{\partial \vec{D}}{\partial t}$$

Perfect Electric Conductor

Perfect Electric Conductor (PEC)

Perfect electric conductor (PEC): $\sigma = \infty$

Fields in PEC vanish.

- 1 From Ohm's law:

$$\vec{J} = \sigma \vec{E}$$

The fact that $\vec{J}$ is finite leads to the conclusion that $\vec{E} = 0$;

- 2 From physics: $\sigma = \infty$ means no resistance to electron motion: electrons repel each other to distribute on the outer surface of PEC.

Gauss's law:

$$\oint \vec{D} \cdot d\vec{S} = Q = 0 \quad \text{inside the PEC.}$$

$$\Rightarrow \vec{D} = \vec{E} = 0$$

$$V_{ab} = - \int_a^b \vec{E} \cdot d\vec{l} = 0 \rightarrow \text{PEC is an equipotential medium.}$$



HOWARD
UNIVERSITY

Imperfect Electric Conductor

Imperfect (Real) Electric Conductor

Potential difference in the direction of the current flow:

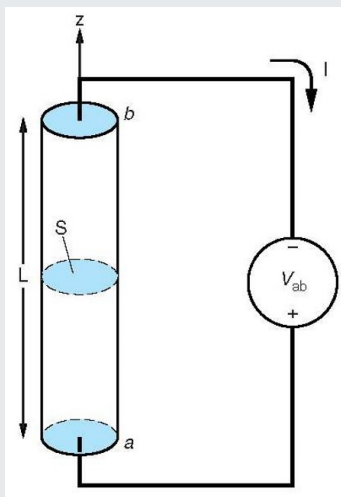


Figure: Bar with conductivity σ for deriving the circuit-theory form of Ohm's law.

- From circuit theory: $V_{ab} = RI$ (Ohm's law)

- From electromagnetic theory:

Voltage:

$$V_{ab} = V_a - V_b = - \int_b^a \vec{E} \cdot d\vec{l}$$

$$= \int_a^b \vec{E} \cdot d\vec{l} = E_z L$$

$$\Rightarrow \vec{E} = \frac{V_{ab}}{L} \hat{z}$$

Current:

$$I = \int \vec{J} \cdot d\vec{S} = J_z \hat{z} \cdot S \hat{z} = J_z S$$

$$\Rightarrow \vec{J} = \frac{I}{S} \hat{z}$$

Since $\vec{J} = \sigma \vec{E}$,

$$\frac{I}{S} = \sigma \frac{V_{ab}}{L} \Rightarrow \frac{V_{ab}}{I} = \frac{1}{\sigma} \frac{L}{S} = R$$

$$\Rightarrow R = \frac{1}{\sigma} \frac{L}{S} = \rho \frac{L}{S}$$



HOWARD
UNIVERSITY

Imperfect Electric Conductor

Imperfect (Real) Electric Conductor

Potential difference in the direction of the current flow:

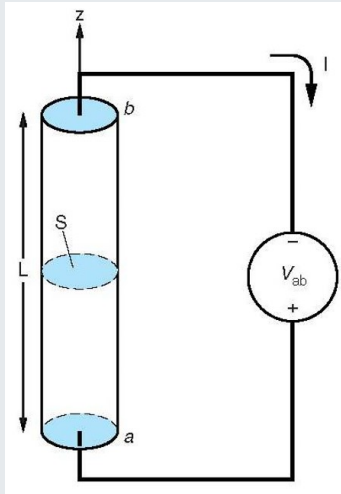


Figure: Bar with conductivity σ for deriving the circuit-theory form of Ohm's law.

■ From circuit theory: $V_{ab} = RI$ (Ohm's law)

■ From electromagnetic theory:

$$R = \frac{1}{\sigma} \frac{L}{S} = \rho \frac{L}{S}$$

$$\text{Resistance} = \frac{1}{\text{Conductivity}} \frac{\text{Length}}{\text{Cross Section}}$$

$$= \text{Resistivity} \frac{\text{Length}}{\text{Cross Section}}$$

■ In general:

$$R = \frac{V_{ab}}{I} = \frac{-\int_b^a \vec{E} \cdot d\vec{l}}{\int \sigma \vec{E} \cdot d\vec{S}}$$



Imperfect Electric Conductor

Example 2.21

Find the resistance between the inner conductive shell with radius a and outer conductive shell with radius b for a length L of a coaxial cable filled with material of conductivity σ .

Solution:

■ Assume charge Q on the inner shell:

For $a \leq \rho \leq b$, applying Gauss's law yields:

$$\vec{E} = \frac{Q}{2\pi\epsilon_0\rho L} \hat{\rho}$$

■ Voltage:

$$V_{ab} = -\int_b^a \vec{E} \cdot d\vec{l} = \int_a^b \vec{E} \cdot d\vec{l} = \int_a^b \frac{Q}{2\pi\epsilon_0\rho L} \hat{\rho} \cdot \hat{\rho} d\rho = \frac{Q}{2\pi\epsilon_0 L} \ln \rho \Big|_a^b = \frac{Q}{2\pi\epsilon_0 L} \ln \left(\frac{b}{a} \right)$$

■ Current:

$$I = \int \vec{j} \cdot d\vec{S} = \int \sigma \vec{E} \cdot d\vec{S} = \int_0^L \int_0^{2\pi} \frac{\sigma Q}{2\pi\epsilon_0\rho L} \hat{\rho} \cdot \hat{\rho} \rho d\phi dz = \frac{\sigma Q}{2\pi\epsilon_0 L} 2\pi L = \frac{\sigma Q}{\epsilon_0}$$

■ Resistance:

$$R = \frac{V_{ab}}{I} = \frac{Q}{2\pi\epsilon_0 L} \ln \left(\frac{b}{a} \right) \frac{\epsilon_0}{\sigma Q} = \frac{1}{2\pi\sigma L} \ln \left(\frac{b}{a} \right)$$



Joule's Law

Joule's Law

- Electric field does work in moving charged particles → Field losses energy.
- Energy loss: Due to the collisions between charged particles and atoms in the material → Generates heat.
- Energy loss per unit time: **Dissipated Power P** and **Dissipated Power Density p** .
In circuit theory, power loss:

$$P = VI$$

Since

$$V = \int_a^b \vec{E} \cdot d\vec{l} = \vec{E} \cdot \hat{l}L$$

$$I = \int \vec{J} \cdot d\vec{S} = \vec{J} \cdot \hat{s}S$$

we have the dissipated power (note $\vec{E}$, $\vec{J}$, $\hat{l}$, and $\hat{s}$ are all in the same direction)

$$P = VI = \vec{E} \cdot \hat{l}L \vec{J} \cdot \hat{s}S = \vec{E} \cdot \vec{J}V$$

and the dissipated power density

$$p = \frac{P}{V} = \vec{E} \cdot \vec{J}$$



Joule's Law

Joule's Law

- **Joule's Law:**

$$p = \vec{E} \cdot \vec{J} = \sigma \vec{E} \cdot \vec{E} = \sigma E^2 \quad \left(\frac{\text{V}}{\text{m}} \cdot \frac{\text{A}}{\text{m}^2} = \frac{\text{W}}{\text{m}^3} \right)$$

- Dissipated Power:

$$P = \int p \, dV = \int \vec{E} \cdot \vec{J} \, dV = \int \sigma E^2 \, dV$$

Joule's Law — A more general derivation.

- Differential force exerted by electric field to move differential charge:

$$d\vec{F} = \vec{E} \, dQ = \vec{E} \rho_V \, dV$$

- Differential work:

$$dW = d\vec{F} \cdot d\vec{l} = \vec{E} \rho_V \, dV \cdot d\vec{l}$$

- Differential dissipated power:

$$dP = \frac{dW}{dt} = \vec{E} \rho_V \, dV \cdot \frac{d\vec{l}}{dt} = \vec{E} \rho_V \, dV \cdot \vec{u} = \vec{E} \cdot (\rho_V \vec{u}) \, dV = \vec{E} \cdot \vec{J} \, dV$$

In the above, $\vec{u}$ denotes the velocity of electron motion.

- As a result:

$$P = \int dP = \int \vec{E} \cdot \vec{J} \, dV = \int \sigma E^2 \, dV \quad \rightarrow \text{Joule's Law}$$



Joule's Law

Example 2.22

Find the dissipated power in a conductive cylinder with a diameter of 10 cm and a height of 1 m. The conductivity is given as $\sigma = 10^{-3}$ S/m and the applied electric field is $\vec{E} = 12\rho\hat{z}$ V/cm, where ρ is in cm.

Solution:

$$\begin{aligned}
 P &= \int \sigma E^2 dV = \sigma \int E^2 dV \\
 &= 10^{-5} \text{ S/cm} \times \int_0^{100 \text{ cm}} \int_0^{2\pi} \int_0^{5 \text{ cm}} (12\rho)^2 \rho d\rho d\phi dz \\
 &= 144 \times 10^{-5} \frac{\rho^4}{4} \Big|_0^5 \times 100 \times 2\pi \\
 &= 45\pi \text{ (W)} \approx 141.4 \text{ (W)}
 \end{aligned}$$



Dielectrics

Dielectrics

- Conductors: motion of free electrons generates currents: $\vec{J} = \sigma\vec{E}$
- Dielectrics: bound charges, very few free charges

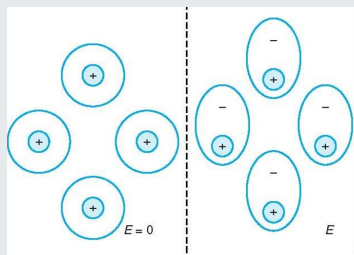


Figure: A collection of atoms before and after application of an electric field shows the formation of aligned dipoles.

- Under the application of an external field $\vec{E}$, the material is **polarized** by the electric field.
- The degree and direction of the alignment is described as

$$\vec{P} = \chi_e \epsilon_0 \vec{E}$$

Polarization Vector = (Electric Susceptibility "Chi") $\times$ (Electric Flux Density)

Electric susceptibility (χ_e): How easily/difficult the material can be polarized.



Dielectrics

Dielectrics

- Total lines of electric flux in the material:

$$\vec{D} = \epsilon_0 \vec{E} + \vec{P} = (1 + \chi_e) \epsilon_0 \vec{E} = \epsilon_r \epsilon_0 \vec{E}$$

- The **constitutive relation**:

$$\vec{D} = \epsilon_r \epsilon_0 \vec{E} = \epsilon \vec{E}$$

where $\epsilon_r = 1 + \chi_e$.

- ϵ : Permittivity of the material;
- ϵ_r : Relative Permittivity of the material;
- If $\epsilon_r = \epsilon_r(\vec{E})$: Nonlinear material

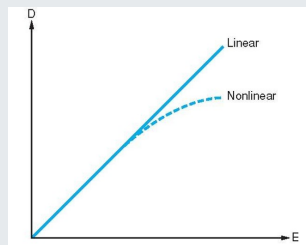


Figure: The $\vec{D}$ versus $\vec{E}$ relation is compared for a linear and a nonlinear material.



Boundary Conditions

Field behavior

- 1 Inside conductors: $\vec{J} = \sigma \vec{E}$

Inside dielectrics: $\vec{D} = \epsilon \vec{E}$

- 2 At the boundaries of dielectric–dielectric, conductor–dielectric: **Boundary Condition**

The conditions that fields must satisfy across the boundary of two materials.



Boundary Conditions

Boundary Condition for Tangential Field Components:

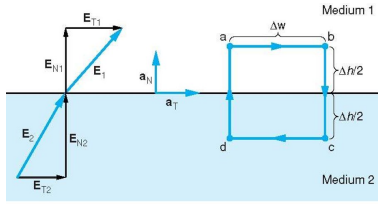


Figure: Boundary condition for a line integral.

- Apply the Kirchhoff's voltage law (KVL):

$$\oint_C \vec{E} \cdot d\vec{l} = 0$$

where C is the contour chosen as

$$C: a \rightarrow b \rightarrow c \rightarrow d \rightarrow a$$

As a result,

$$\oint_C \vec{E} \cdot d\vec{l} = \int_a^b + \int_b^c + \int_c^d + \int_d^a = 0$$

- For each line segment:

$$\int_a^b \vec{E} \cdot d\vec{l} = E_{1t} \cdot \Delta W$$

$$\int_b^c \vec{E} \cdot d\vec{l} = -E_{1n} \cdot \frac{\Delta h}{2} - E_{2n} \cdot \frac{\Delta h}{2} = -(E_{1n} + E_{2n}) \cdot \frac{\Delta h}{2}$$

$$\int_c^d \vec{E} \cdot d\vec{l} = -E_{2t} \cdot \Delta W$$

$$\int_d^a \vec{E} \cdot d\vec{l} = E_{1n} \cdot \frac{\Delta h}{2} + E_{2n} \cdot \frac{\Delta h}{2} = (E_{1n} + E_{2n}) \cdot \frac{\Delta h}{2}$$

- Now, examine the condition right on the boundary by setting $\Delta h \rightarrow 0$:

$$\int_b^c \vec{E} \cdot d\vec{l} = \int_d^a \vec{E} \cdot d\vec{l} = (E_{1n} + E_{2n}) \cdot \frac{\Delta h}{2} = 0$$

- KVL leads to:

$$(E_{1t} - E_{2t})\Delta W = 0 \Rightarrow \boxed{E_{1t} = E_{2t}}$$

→ **The tangential component of the electric field intensity is continuous across the boundary of two media.**

$$\hat{n} \times \vec{E}_1 = \hat{n} \times \vec{E}_2$$



Boundary Conditions

Boundary Condition for Normal Flux Components:

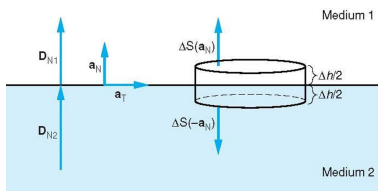


Figure: Boundary condition for Gauss's law.

- Apply Gauss's law:

$$\oint_S \vec{D} \cdot d\vec{S} = Q_{enc}$$

where S is the closed surface chosen as the "pillbox".

Note: $\vec{D} = D_n \hat{n} + D_t \hat{t}$.

As a result,

$$\oint_S \vec{D} \cdot d\vec{S} = \int_{top} + \int_{bottom} + \int_{side}$$

- For each surface integral:

$$\int_{top} \vec{D} \cdot d\vec{S} = D_{1n} \hat{n} \cdot \hat{n} \Delta S$$

$$\int_{bottom} \vec{D} \cdot d\vec{S} = D_{2n} \hat{n} \cdot (-\hat{n}) \Delta S$$

$$\int_{side} \vec{D} \cdot d\vec{S} = \int_{side1} D_{1t} \hat{t} \cdot \hat{\rho} dS + \int_{side2} D_{2t} \hat{t} \cdot \hat{\rho} dS$$

- Now, examine the condition right on the boundary by setting $\Delta h \rightarrow 0$:

$$\int_{side} \vec{D} \cdot d\vec{S} = 0$$

- Gauss's law leads to:

$$(D_{1n} - D_{2n})\Delta S = Q_{enc} = \rho_S \Delta S$$

$$\Rightarrow \boxed{D_{1n} - D_{2n} = \rho_S}$$

→ **The discontinuity of the normal component of the electric flux density is supported by the surface charge density.**

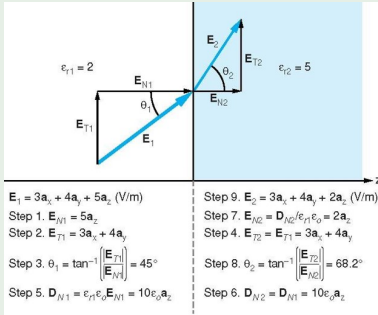
$$\hat{n} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_S$$



Boundary Conditions

Example 2.24

Find $\vec{E}_2$ and θ_2 in medium 2 where $\vec{E}_1 = 3\hat{x} + 4\hat{y} + 5\hat{z}$ V/m.



Solution:

- Tangential component of electric field is continuous:

$$E_{1t} = E_{2t}$$

$$3\hat{x} + 4\hat{y} = E_{2t}$$

- Normal component of electric flux is continuous (since there is no surface charge):

$$D_{1n} = D_{2n}$$

$$\epsilon_1 E_{1z} = \epsilon_2 E_{2z}$$

$$\epsilon_0 \epsilon_{r1} E_{1z} = \epsilon_0 \epsilon_{r2} E_{2z}$$

$$2 \times 5 = 5 \times E_{2z} \Rightarrow E_{2z} = 2 \text{ V/m}$$

- We have

$$\vec{E}_2 = 3\hat{x} + 4\hat{y} + 2\hat{z} \text{ V/m}$$

- The angle made between $\vec{E}_2$ and the normal direction:

$$\theta_2 = \tan^{-1} \frac{|E_{2t}|}{|E_{2n}|} = \tan^{-1} \frac{5}{2} = 68.2^\circ$$



Boundary Conditions

Boundary Conditions on Dielectric–Dielectric Boundaries

- Tangential Field Component:

$$\hat{n} \times \vec{E}_1 = \hat{n} \times \vec{E}_2$$

- Normal Flux Component:

$$\hat{n} \cdot (\vec{D}_1 - \vec{D}_2) = \rho_s$$

Boundary Conditions on Dielectric–PEC Boundaries

- Tangential Field Component:

$$\hat{n} \times \vec{E}_1 = 0$$

- Normal Flux Component:

$$\hat{n} \cdot \vec{D}_1 = \rho_s$$



Boundary Value Problem (BVP)

- Solve for the field distribution everywhere given the information of the source distribution and the boundary condition (B.C.).

Poisson's Equation & Laplace's Equation

- From Gauss's law:

$$\nabla \cdot \vec{D} = \rho_V$$

the constitutive relation

$$\vec{D} = \epsilon \vec{E}$$

and assume ϵ is a constant, we have

$$\nabla \cdot \vec{E} = \frac{\rho_V}{\epsilon}$$

- Expressing the electric field as

$$\vec{E} = -\nabla V$$

yields **Poisson's Equation**:

$$\nabla \cdot \nabla V = \boxed{\nabla^2 V = -\frac{\rho_V}{\epsilon}}$$

where ∇^2 is the **Laplacian Operator**.

- Source-free case: $\rho_V = 0$

$$\boxed{\nabla^2 V = 0} \rightarrow \text{Laplace's Equation}$$

- Laplacian Operator:

Rectangular Coordinate System:

$$\nabla^2 V = \frac{\partial^2 V}{\partial x^2} + \frac{\partial^2 V}{\partial y^2} + \frac{\partial^2 V}{\partial z^2}$$

Cylindrical Coordinate System:

$$\nabla^2 V = \frac{1}{\rho} \frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) + \frac{1}{\rho^2} \frac{\partial^2 V}{\partial \phi^2} + \frac{\partial^2 V}{\partial z^2}$$

Spherical Coordinate System:

$$\nabla^2 V = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial V}{\partial \theta} \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 V}{\partial \phi^2}$$



Boundary Value Problem (BVP)

Objective

- Solve Poisson's equation

$$\nabla^2 V = -\frac{\rho_V}{\epsilon}$$

together with source distribution ρ_V

and boundary condition

$$\begin{aligned} V &= V_o, & \text{on } \Gamma_{\text{Dirichlet}} \\ \frac{\partial V}{\partial n} &= \Phi_o, & \text{on } \Gamma_{\text{Neumann}} \end{aligned}$$

for electrostatic potential to find electric field distribution and/or other physical quantities of interest, such as

$$R = \frac{-\int_b^a \vec{E} \cdot d\vec{l}}{\int \sigma \vec{E} \cdot d\vec{S}}.$$



Boundary Value Problem (BVP)

Example 2.25

Find the resistance of the following parallel plates, assuming $V(z = 0) = 0$ (V), and $V(z = d) = V_d$ (V).

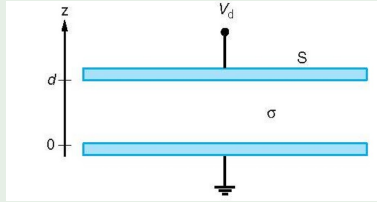


Figure: Cross section of a block of material for which resistance is to be calculated.

Solution:

- No source: $\rho_V = 0$

$$\nabla^2 V = 0.$$

- If neglect the edge effect (field distribution around the edges): $V = V(z)$

$$\frac{\partial^2 V(z)}{\partial z^2} = 0$$

- Integrate once over z :

$$\frac{\partial V(z)}{\partial z} = A$$

Integrate the second time over z :

$$V(z) = Az + B$$

where A and B are unknown constants.



Boundary Value Problem (BVP)

Example 2.25

Find the resistance of the following parallel plates, assuming $V(z = 0) = 0$ (V), and $V(z = d) = V_d$ (V).

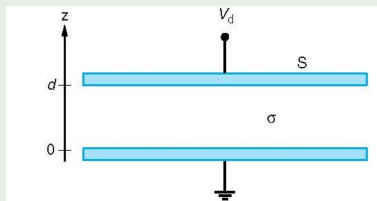


Figure: Cross section of a block of material for which resistance is to be calculated.

- To determine the unknown constants A and B :

Math the boundary conditions:

$$V(z = 0) = 0$$

$$V(z = d) = V_d$$

- Substituting the above boundary conditions into the general solution

$$V(z) = Az + B$$

yields

$$B = 0$$

$$Ad + B = V_d$$

- As a result

$$A = \frac{V_d}{d}$$

$$B = 0$$



Boundary Value Problem (BVP)

Example 2.25

Find the resistance of the following parallel plates, assuming $V(z=0) = 0$ (V), and $V(z=d) = V_d$ (V).

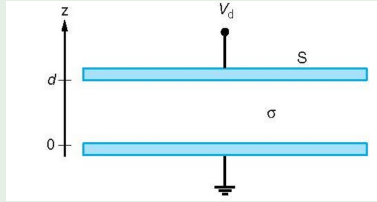


Figure: Cross section of a block of material for which resistance is to be calculated.

- The electrostatic potential:

$$V(z) = \frac{V_d}{d}z$$

- The electric field can be found as

$$\vec{E} = -\nabla V(z) = -\frac{\partial V}{\partial z}\hat{z} = -\frac{V_d}{d}\hat{z}$$

- Current:

$$\begin{aligned} I &= \int \vec{J} \cdot d\vec{S} = \int \sigma \vec{E} \cdot d\vec{S} \\ &= -\frac{\sigma V_d}{d}\hat{z} \cdot S(-\hat{z}) = \frac{\sigma S}{d}V_d \end{aligned}$$

- Voltage:

$$V_d$$

- Resistance:

$$R = \frac{V_d}{I} = \frac{1}{\sigma} \frac{d}{S}$$



Boundary Value Problem (BVP)

Example 2.26

Find the potential distribution between the concentric spheres with potential difference V_a and charge density $\rho_V = \frac{\rho_0}{r}$ in between the the spherical shells.

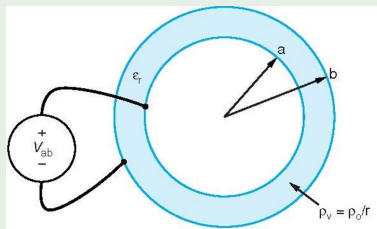


Figure: Concentric conductive spheres with a charge distribution between them.

Solution:

- Due to symmetry:

$$V = V(r)$$

- Poisson's equation $\nabla^2 V = -\rho_V/\epsilon$ in spherical coordinate system becomes

$$\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial V(r)}{\partial r} \right) = -\frac{\rho_0}{\epsilon_0 \epsilon_r r}$$

- Multiply both sides by r^2 :

$$\frac{\partial}{\partial r} \left(r^2 \frac{\partial V(r)}{\partial r} \right) = -\frac{\rho_0}{\epsilon_0 \epsilon_r} r$$

Integrate once over r :

$$r^2 \frac{\partial V(r)}{\partial r} = -\frac{\rho_0}{2\epsilon_0 \epsilon_r} r^2 + A$$



Boundary Value Problem (BVP)

Example 2.26

Find the potential distribution between the concentric spheres with potential difference V_a and charge density $\rho_V = \frac{\rho_o}{r}$ in between the the spherical shells.

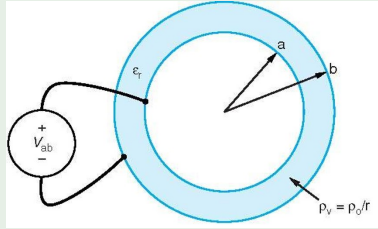


Figure: Concentric conductive spheres with a charge distribution between them.

Solution:

- Divide both sides by r^2 :

$$\frac{\partial V(r)}{\partial r} = -\frac{\rho_o}{2\epsilon_o\epsilon_r} + \frac{A}{r^2}$$

Integrate over r again:

$$V(r) = -\frac{\rho_o}{2\epsilon_o\epsilon_r}r - \frac{A}{r} + B$$

where A and B are unknown constants.

- To determine A and B , match the boundary conditions:

$$V(r=a) = -\frac{\rho_o a}{2\epsilon_o\epsilon_r} - \frac{A}{a} + B = V_a$$

$$V(r=b) = -\frac{\rho_o b}{2\epsilon_o\epsilon_r} - \frac{A}{b} + B = 0$$



HOWARD
UNIVERSITY

Boundary Value Problem (BVP)

Example 2.26

Find the potential distribution between the concentric spheres with potential difference V_a and charge density $\rho_V = \frac{\rho_o}{r}$ in between the the spherical shells.

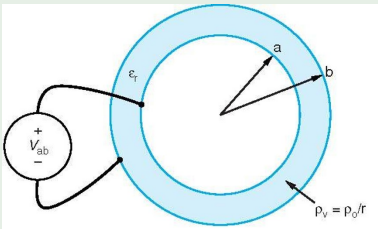


Figure: Concentric conductive spheres with a charge distribution between them.

Solution:

- Subtracting the above two equations yields

$$\frac{a-b}{ab}A + \frac{\rho_o}{2\epsilon_o\epsilon_r}(b-a) = V_a$$

The unknown constants can be found as

$$A = \frac{ab}{b-a} \left[\frac{\rho_o}{2\epsilon_o\epsilon_r}(b-a) - V_a \right]$$

$$= \frac{\rho_o ab}{2\epsilon_o\epsilon_r} - \frac{ab}{b-a} V_a$$

$$B = \frac{\rho_o b}{2\epsilon_o\epsilon_r} + \frac{A}{b} = \frac{\rho_o b}{2\epsilon_o\epsilon_r} + \frac{\rho_o a}{2\epsilon_o\epsilon_r} - \frac{a}{b-a} V_a$$

$$= \frac{\rho_o}{2\epsilon_o\epsilon_r}(a+b) - \frac{a}{b-a} V_a$$

- The electrostatic potential:

$$V = \frac{\rho_o}{2\epsilon_o\epsilon_r} \left[-r - \frac{ab}{r} + a + b \right] + \frac{a}{b-a} V_a \left[\frac{b}{r} - 1 \right]$$



HOWARD
UNIVERSITY

Capacitance

Parallel-Plate Capacitor

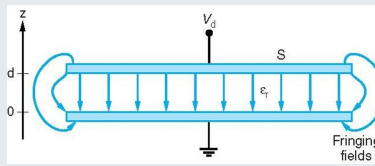


Figure: Cross section of a parallel-plate capacitor.

- Applying the potential difference V_d between two parallel plates results in the accumulation of $+Q$ and $-Q$ charges on the two plates.
- **Capacitance:** The amount of charges that accumulates as a function of potential difference.

$$C = \frac{Q}{V}$$

Units: Farad = F = C/V.

- **Capacitor:** A device used to store charges, hence, electrical energy.

Typical values: μF (10^{-6} F), pF (10^{-12} F)



Capacitance

Calculation of Capacitance

Capacitance depends on:

- 1 Conductor-dielectric configuration (shape, etc.);
- 2 Dielectric's permittivity (ϵ).

Two methods:

- 1 Q-Method: Given Q , find V , and $C = Q/V$;
- 2 V-Method: Given V , find Q , and $C = Q/V$.

Q-Method

- 1 Assume a charge $+Q$ on plate a , and $-Q$ on plate b ;
- 2 Solve for $\vec{E}$ with Coulomb's law, Gauss's law, or boundary condition;
- 3 Solve for the potential difference V ;
- 4 Find the capacitance $C = Q/V$.

V-Method

- 1 Assume V between plates;
- 2 Find $\vec{E}$, then $\vec{D}$, using Laplace's equation;
- 3 Find ρ_s , then Q , on each plate using the boundary condition $D_n = \rho_s$;
- 4 Find the capacitance $C = Q/V$.



Capacitance

Example 2.27

Find the capacitance for the parallel-plate capacitor using the Q -method.

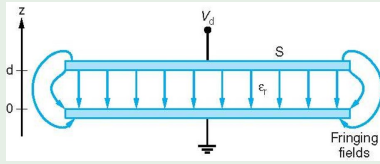


Figure: Cross section of a parallel-plate capacitor.

Relation between R and C

Note: for the same structure:

$$R = \frac{1}{\sigma} \frac{d}{S}, \quad C = \frac{\epsilon S}{d}$$

So we have the relation:

$$RC = \frac{\epsilon}{\sigma}$$

This is a general relation between R and C .

Solution:

- 1 Place $\pm Q$ on the plates;
- 2 The surface charge density: $\rho_s = Q/S$;
- 3 The electric flux density can be expressed as:

$$\vec{D} = \frac{Q}{S}(-\hat{z})$$

and the electric field intensity:

$$\vec{E} = \frac{\vec{D}}{\epsilon} = -\hat{z} \frac{Q}{\epsilon_0 \epsilon_r S}$$

- 4 The potential:

$$V = - \int_b^a \vec{E} \cdot d\vec{l} = - \int_0^d \frac{-Q}{\epsilon_0 \epsilon_r S} dz = \frac{Qd}{\epsilon_0 \epsilon_r S}$$

- 5 The capacitance:

$$C = \frac{Q}{V} = \frac{\epsilon_0 \epsilon_r S}{d}$$



Capacitance

Example 2.28

Find the capacitance of a coaxial cable using the V -method.

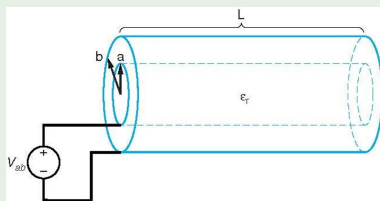


Figure: Coaxial capacitor.

Solution:

1. Apply voltage V_{ab} .
Neglect fringing fields and assume $V = V(\rho)$.
2. Solve Laplace's equation

$$\nabla^2 V = 0$$

In cylindrical coordinate system:

$$\frac{\partial}{\partial \rho} \left(\rho \frac{\partial V}{\partial \rho} \right) = 0$$

Solve the equation:

$$\rho \frac{\partial V}{\partial \rho} = A, \quad \Rightarrow \quad V(\rho) = A \ln \rho + B$$

3. Match the boundary condition:

$$V(a) = V_{ab}, \quad V(b) = 0$$

to get

$$A \ln a + B = V_{ab}$$

$$A \ln b + B = 0.$$



Capacitance

Example 2.28

Find the capacitance of a coaxial cable using the V -method.

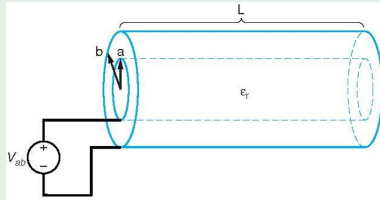


Figure: Coaxial capacitor.

Solution:

The unknown constants are found as

$$A = -\frac{V_{ab}}{\ln(b/a)}$$

$$B = -A \ln b = \frac{V_{ab} \ln b}{\ln(b/a)}.$$

4. The potential distribution:

$$V(\rho) = -\frac{V_{ab} \ln \rho}{\ln(b/a)} + \frac{V_{ab} \ln b}{\ln(b/a)} = \frac{\ln(b/\rho)}{\ln(b/a)} V_{ab}$$

5. Next, find $\vec{E}$:

$$\begin{aligned} \vec{E} &= -\nabla V = -\frac{\partial V}{\partial \rho} \hat{\rho} = -\hat{\rho} \frac{V_{ab}}{\ln(b/a)} \frac{\partial}{\partial \rho} \ln\left(\frac{b}{\rho}\right) \\ &= -\hat{\rho} \frac{V_{ab}}{\ln(b/a)} \left(-\frac{1}{\rho}\right) = \hat{\rho} \frac{V_{ab}}{\rho \ln(b/a)} \end{aligned}$$

The electric flux:

$$\vec{D} = \epsilon_0 \epsilon_r \vec{E} = \hat{\rho} \frac{\epsilon_0 \epsilon_r V_{ab}}{\rho \ln(b/a)}$$



Capacitance

Example 2.28

Find the capacitance of a coaxial cable using the V -method.

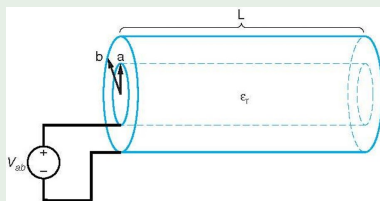


Figure: Coaxial capacitor.

Solution:

6. Charge density on the inner conductor:

$$\rho_S = \hat{n} \cdot \vec{D} \Big|_{\rho=a} = \hat{\rho} \cdot \vec{D} \Big|_{\rho=a} = \frac{\epsilon_0 \epsilon_r V_{ab}}{a \ln(b/a)}$$

1. Total charge on the inner conductor:

$$Q = \rho_S S = \frac{\epsilon_0 \epsilon_r V_{ab}}{a \ln(b/a)} \times 2\pi a L = \frac{2\pi L \epsilon_0 \epsilon_r V_{ab}}{\ln(b/a)}$$

7. The capacitance:

$$C = \frac{Q}{V} = \frac{2\pi L \epsilon_0 \epsilon_r}{\ln(b/a)}$$



Electrostatic Potential Energy

Electrostatic Potential Energy

- Moving like charges together requires work.
- In other words, a collection of charges poses potential energy.
- Considering the work taken to assemble charges:

$$W_E = \frac{1}{2} \int \vec{D} \cdot \vec{E} dV = \frac{1}{2} \int \epsilon_0 \epsilon_r E^2 dV$$

— Electrostatic Potential Energy

- Energy density:

$$\frac{1}{2} \vec{D} \cdot \vec{E} \quad \text{Units: J/m}^3$$

- For a parallel-plate capacitor:

$$W_E = \frac{1}{2} \epsilon_0 \epsilon_r E^2 Sd, \quad E = \frac{V}{d}, \quad C = \frac{\epsilon_0 \epsilon_r S}{d}$$

we have

$$W_E = \frac{1}{2} \epsilon_0 \epsilon_r \left(\frac{V}{d} \right)^2 Sd = \frac{1}{2} \frac{\epsilon_0 \epsilon_r S}{d} V^2 = \frac{1}{2} CV^2$$

— energy stored in a capacitor

- This holds true for any capacitor.



HOWARD
UNIVERSITY

Electrostatic Potential Energy

Example 2.29

Parallel-plate capacitor with $S = 4 \text{ m}^2$, $d = 0.01 \text{ m}$, $\epsilon_r = 10$, $\sigma = 10^{-8} \text{ S/m}$. Apply potential $V = 12 \text{ V}$. Find:

- 1 Stored energy W_E ;
- 2 Dissipated power P .

Solution:

- 1 Find the capacitance:

$$C = \frac{\epsilon_0 \epsilon_r S}{d} = \frac{8.854 \times 10^{-12} \times 10 \times 4}{10^{-2}} = 35 \text{ nF}$$

- 2 Potential energy:

$$W_E = \frac{1}{2} CV^2 = \frac{1}{2} \times 35 \times 10^{-9} \times 12^2 = 210 \text{ nJ}$$

- 3 Dissipated power:

$$\begin{aligned} P &= \sigma E^2 \cdot Sd = \sigma \left(\frac{V}{d} \right)^2 \cdot Sd \\ &= 10^{-8} \times \left(\frac{12}{0.01} \right)^2 \times 4 \times 0.01 = 0.58 \text{ mW} \end{aligned}$$



HOWARD
UNIVERSITY